

Introduction

Recognition is a process fundamental to humans, allowing us to judge that a current event is similar to one or more past events. The goal of this research is the formulation of a model of the most important processes that are used when recognition memory decisions are made for long-term event memories (also termed episodic memories). With this in mind, the studies modeled use a distracting task after list study and prior to test to clear memory traces held in short-term memory. The research uses data from two studies with novel designs to test recognition memory. The model to explain that data (and also to explain results from other studies) is based on the 'Retrieving Effectively from Memory (REM) model (Shiffrin & Steyvers, 1997, 1998). The 1997 paper presented a simple version of the model followed by more complex extensions. The simplest version will be termed bREM (standing for basic REM). bREM is a very simple three parameter model that has proved enormously successful in predicting the major trends in recognition studies, even when using a fixed set of parameter values for all studies and conditions (see Mohamed, Meyer-Grant, & Shiffrin, submitted). Nonetheless, bREM is known to be missing quite a few processes that are important for recognizing, and the present research is designed to fill in some of the important gaps.

We begin by outlining the core mechanisms of bREM. bREM is an example of the most commonly used account of recognition by familiarity. Familiarity is almost always assumed to be produced by global matching processes (see discussion in Gillund & Shiffrin, 1984). Global matching is a term used to describe retrieval that uses similarity to match the test item to all memory traces in parallel, combining the comparisons to produce a 'feeling of familiarity' that is used to decide whether the test item is old or new. This type of recognition modeling is distinct from recognition decisions based on recall or retrieval of individual traces (an important component of present modeling).

bREM is also a variant of the most commonly used way of analyzing and modeling recognition studies: signal detection theory (SDT). [REFERENCE] As described below, both bREM and SDT posit distributions of decision variable 'strengths'. The strengths are used to give either an 'old' (an item studied) or a 'new' (an item not studied) decision. Both have a distribution of strengths for foils (test items not studied) and another for targets (test items studied), with higher variance for the target distribution. Both have a decision criterion such that strengths below the criterion produce a new response and strengths above produce an old response. The distributions in SDT are assumed to be Gaussian, whereas the distributions in bREM are enormously skewed toward large values, especially for targets. In SDT the decision criterion, and possibly other parameters, are varied with the experimental conditions. In bREM the decision criterion and most other parameters do not vary with the experimental conditions. We aim to produce an extended model based on bREM rather than SDT because bREM and its extension are causal mechanistic models positing the basis for the strengths that are used to

decide. Although the extended REM model for the present studies will be quite complex, it will still be simpler than an extended SDT model because it will use (mostly) the same decision criterion for different conditions, and will use fewer parameter values that differ among conditions.

Discussing bREM allows us to show how a few critical processes are sufficient to account for a surprising number of observed trends in recognition studies. It also allows us to set the stage for the later elaborations of the model that are layered on top of bREM—including the use of context, the dynamics of context change, differentiation of feature types, the storage of new traces at test, the updating of existing traces at test, and the contribution of recall to recognition decisions. This discussion of recognition processes missing from bREM leads naturally to the rationale for using REM as the basis for formulating a (more) general account of recognition memory. The missing processes also motivate the design of the two new studies that are presented in the next section.

bREM:

A] Representations of items and traces

Every item and every memory trace of an item is represented by a vector of feature values (arbitrarily set to 20 by Shiffrin & Steyvers, 1997). For each item, each value is chosen independently from a geometric distribution with values 1, 2, etc. The values represent base rate frequencies of that feature in the environment of in the experimental context, or both. 1 represents the most common value of that feature and higher numbers are successively less frequent. The decrease in values is geometric with parameter g (set to 0.4 in REM 1997 and 0.3 in the present research, though these values make little difference in the pattern of predictions for the present studies).

B] Storage of memory traces

Let us assume N items are studied. When an item is studied each feature value is stored in a memory trace for that item with probability u or not stored at all (the number 0 is used to represent a missing value in the trace). If stored the value is stored correctly (i.e. copied into the trace) with probability c ; with probability $1 - c$ the value stored is a random choice from the geometric distribution. In REM 1997 u was set to (the equivalent of) 0.4 and the present modeling uses 0.4 as well (though again this value is not critical for producing the patterns of data in the present studies). REM 1997 was used to predict studies using words as stimuli, and set $c = 0.7$. The present studies used pictures, and pictures of the type used in these studies and most others (common objects taken from the set used and studied by Brady Konkle Alvarez and Oliva (2008) are remembered better than words. Thus in the present modeling c for pictures was

set to 0.85 for Experiment 1; it was set to a lower value of 0.7 for Experiment 2 (for reasons inherent in the design of Experiment 2, as discussed later). The result of storage is N incomplete and partially correct memory traces.

C] Recognition decisions

For the studies modeled (in this research and most other research) recognition testing requires a decision whether the test item had been on the studied list or not. Half the test items were on the list (termed ‘target’, or ‘old’) and half the test items were not on the list (termed ‘foil’, or ‘new’). In bREM the 20 values of the item presented for test are matched in parallel to the twenty incomplete and error prone values in each of the N traces. Each feature comparison produces a likelihood ratio that the value was produced by study of the test item vs study of something else (matching values produce likelihood ratios greater than 1; mismatching values produce likelihood ratios less than 1). When a trace value is missing (value 0) it is ignored. The product of the remaining feature likelihood ratios give the likelihood ratio of that trace. The average of the N trace likelihood ratios gives the odds that the test item was on the list. If the odds is greater than 1.0 an old response is made, otherwise a new response is made. The equations giving the odds are based on an ideal Bayesian decision derivation (for details, see Shiffrin & Steyvers, 1997). bREM is an extremely simple model with just three parameters, and was not meant to be a reasonably complete model of recognition.

D] What bREM predicts

Given its simplicity and the many processes known to contribute to recognition that are missing from bREM it is remarkable that this model correctly predicts the main findings from recognition studies. In 1997 bREM was applied to studies using words as stimuli and quite accurately predicted effects of list length, list strength, study time, word frequency, and the amount of extra estimated variance of targets vs foils (termed zROC). CITE MORE THINGS HERE WHERE REM WORKED. Looking ahead to the present modeling, it is worth noting that predicting ‘list strength’ required strengthening a trace, and that required recalling that trace, so even bREM had a role for recall in recognition.

This year Mohamed et al (submitted) greatly extended the range of conditions to which bREM was applied: 12 item and 24 item lists of word and pictures and mixed lists of words and pictures length were studied. One group (a control) carried out single item old/new recognition as was predicted by bREM in 1997. New paradigms were used for the remaining groups: Two items were always presented for test, both old, one old and one new, or both new. In 2AFC the participant was asked to choose the item more likely to be old, even when both might be old or might be new. In 4WC the participant classified the two items as both old, left old and right new, left new and right old, or both new. Another group repeated all these conditions but judged items

to be new rather than old (which prior research showed produced a different pattern of responses). Over the three experiments there were over 500? different conditions with distinct results. bREM was used to predict every condition using the u and g parameter values from 1997, and using one c parameter value for words and another for pictures (because the standard sets of pictures were known to produce better performance than that for the standard set of words that were used). Applying bREM to tests of two items was straightforward: An odds is calculated for each item; in 2AFC the item with larger odds is chosen; in 4WC, each item's odds is used to classify the pair, using the usual assumption that an old response is given for odds greater than 1.0. The patterns of correct and error responses for the 500+? conditions were correctly predicted perfectly and the predictions were in most cases quite close quantitatively. Mohamed et al (submitted) concluded that such excellent predictions for such a large range of conditions by such a simple model (without tailoring parameter values to each condition) was unlikely to be an accident and suggested that the few core processes in REM were likely fundamental to the way humans recognize. It was these results and conclusions that led us to use bREM as a basis for further model development aimed to fill in many of the processes missing from bREM. The two studies to be described shortly were designed to provide data extending paradigms previously investigated, and to point the way to implement the extensions to the basic model.

E] Augmented REM

When modeling the data from the two new studies, we do not aim to include all important processes missing from bREM. That would be too large a step and lead to a model far too complex to evaluate or understand. However there are several processes known to be important for recognizing will be included in the new modeling, processes informed by the new studies. These are described in turn.

1. *Context and Content Features:*

bREM has only one type of features. Every model of memory finds it necessary to differentiate different types of features.

Content Features: Term these T features. These features differentiate one item from another. Thus for visually presented words such features might include the shape of the letters, the spelling of the word, and the semantic content (i.e. meaning). For pictures, we can again distinguish features, ranging from physical features like color and shape to high level meaning features (e.g. “a beach scene”; “a basketball game”). Cox and Shiffrin (2017) distinguished content feature types in a REM based model that focused mainly on the different ‘arrival’ times of the different feature types. Much research over the years not aimed at dynamics and not based on REM has taken feature differences into account. We cite only one recent publication by Guitard, Saint-Aubin, Reid, and Jamiesn (2025). The present research and modeling does not

manipulate content feature types and does not consider the dynamics of retrieval of different feature types, so these extensions of bREM are left for future extensions. Thus the present extensions of bREM treat all content features as one type, denoted T.

Context Features: Context features are needed to focus retrieval on a specific set of memory traces. That these are critical for storing traces and retrieving them from memory has been a staple of memory theorizing forever (e.g. see James, 1890, chapter XVI; Bower, 1972; Tulving, 1974). There are an almost uncountable number of event traces stored in a human's lifetime, and some means is needed to focus retrieval on just a small subset of these, for example on traces of items studied in the most recent list.

The distinction between content and context is not precise, primarily because items can be associated together (see the discussion of associations later in this report). E.g. the 'Temporal Context Model' (TCM; Howard & Kahana, 2002) defines local context for a list item in terms of the items studied earlier on that list. Alternatively, if one elaborated a model to include associations between items as features, or assume associations between traces, then one might consider the role of items studied earlier as content (or as both as in Murnane, Phelps & Malmberg, 1999).

In any event, context can be considered to be features identifying the situation at the time of storage, features that in most studies are not explicitly identified or manipulated. Such features might represent the physical environment and the internal state of a participant.

Use of Content and Context Features to Retrieve:

When retrieval takes place one would expect that both context and content features would be used together to probe memory (as in Cox & Shiffrin, 2017). However, Shiffrin and Steyvers (1997) proposed a simpler-to-implement approach in which context features alone are used in a first stage of retrieval to determine which memory traces are activated and passed to a second stage. The traces activated are those with context features sufficiently similar to the probe context features. Each trace activated is passed to a second stage. The second stage uses content features only, as in bREM. In this approach context is used as a 'filter' in stage one to activate traces and pass them to the second stage only if their context is similar enough to the probe context. We have observed that the one stage and two stage approaches usually give similar results (at least for predicting choice probabilities). Thus the present research uses the two stage approach. (The two stage model should not be considered two successive stages each taking time, but rather a convenient approximation to a simultaneous use of context and content.)

2. Changes of context as an experiment continues:

Context can be considered to change within list as the items are successively studied (as in TCM), to change between list study and test of that list, to change during a period of testing, to change between one list test and study of the next, to change after all the initial set of lists is completed and before a final test of all seen items is given (as in the present studies), and to change during final testing. Many of these changes of context are needed to deal with the data from the present studies.

Changes of context are generally considered critical to explain a good deal of long-term forgetting. Events are stored with current context, context that usually changes until testing occurs. There is a tendency to retrieve using the changed context available at test, resulting in forgetting (see Estes, 1950; Estes & Burke, 1953). A number of studies have shown that use of initially stored context to retrieve produces better memory than the use of changed context to retrieve (e.g. Smith & Vela, 2001). The improvements are generally modest in size, likely because the studies manipulate just one context factor when both storage and retrieval use multiple context factors. Murnane, Phelps and Malmberg (1999; the ICE model) show that the benefits of context reconstruction in recognition testing are larger when context and content are stored in a trace in integrated rather than incidental fashion. Bauml, Meixensperger and Hirsch (in press, and prior papers) have shown that much forgetting in cued recall after significant delays can be overcome by testing that allows reconstruction of stored context. Their studies show that such reconstruction does not happen when delay until test is less than an hour. The present studies use testing in a single session shorter than an hour, so the delays are not long enough for us to add context reconstruction to the model.

3. Storage of traces during testing:

Testing an item is an event and like all events gets stored. Thus when a foil is tested and judged new, a test trace could be stored in a manner similar to storage of a studied item. When a test item has an existing trace in a context similar to the test context, and the existing trace is recalled, it is not clear how much information about the test is added to the existing trace (see the next section discussing ‘strengthening’) and how much information is stored in the new trace (see Hintzman, Curran & Oppy, 1992; Malmberg, Holden & Shiffrin, 1994). To explain list-strength effects Shiffrin and Steyvers (1997, 1998) assumed that new test traces would not be stored in most cases when an existing strength was strengthened. It is not clear that such an assumption is necessary: Strengthening an existing trace might lead to less attention and a lesser amount of storage in a new trace. It likely would be hard to distinguish a failure to store a new trace from storage of a weak trace. It is convenient in the present modeling to store every study and test event in the same manner. In addition, recall of an existing trace may occur during study or test, and cause the recalled trace to be strengthened, as described in the next section.

Storage of a new test trace plays a role in producing output interference (OI). OI Storing test traces increases the total number of activated traces, increasing effective list length, and thereby reducing performance. OI has been demonstrated in recognition testing (Criss, Malmberg & Shiffrin, 2011; Malmberg, Criss, Gangwani, Shiffrin, 2012; Malmberg, Lehman, Annis, Criss & Shiffrin, 2014). These authors modeled the findings of OI by including in the model storage of test traces (and strengthening of existing traces, as described in the next section). However, list length effects tend to be small in magnitude (see Osth & Dennis, 2015), and there are other factors that might play a larger role in producing OI. Osth, Jansson, Dennis and Heathcote (2018) tested several models and concluded context drift was the most important factor producing OI, but these authors failed to consider the role of strengthening traces (see next).

4. *Strengthening existing traces:*

Raijmakers and Shiffrin (1980, 1981) laid out the role of strengthening existing traces in recall tasks (referred to as incrementing). The sampling was assumed proportional to the strengths of the traces. Proportional sampling might or might not be the best choice but was chosen for simplicity and convenience, What is more important is that the choice be probabilistic. Sampling the strongest trace might work reasonably well in practice, but is hard to justify (for further discussion see Shiffrin & Steyvers, 1997).

Strengthening of existing traces was the basis of the explanation formed by Shiffrin, Ratcliff and Clark (1990) to explain the list-strength effect (LSE). LSE referred to findings that storing some items more strongly at study helped rather harmed recognition of the remaining items (Ratcliff, Clark & Shiffrin, 1990). The authors argued that strengthening traces made them less similar to other traces, causing ‘differentiation’ and reducing confusion. Indeed differentiation due to differential **study** time is embedded in bREM.

It is important to implement also strengthening of traces that occurs during **testing**. The concept is similar: In parallel with computation of familiarity (odds in bREM) a trace might be recalled. The recalled trace will be augmented by addition of features of the test item so that the trace and test item become more similar. If the trace in question was produced by study of a target being tested, the added features will improve the accuracy of a later test of that item. The strengthening also makes that trace less similar to others (differentiation). By itself this factor will improve recognition of other items yet to be tested, but that is not the only result of strengthening as discussed next.

In most studies there are a good number of false alarms when foils are tested. In such cases (other than the present Experiment 2) any recalled trace will be an existing one that was produced by some previously seen item other than the current foil. Strengthening its trace will add features of the test foil thereby degrading the similarity of the trace to the item that produced

it, and lowering performance when and if that item is later tested. This factor combined with the increased number of traces stored during testing can dominate the gains from differentiation, and produce OI. How much of the OI is due to these two different factors is a quantitative question to be answered by modeling.

5. *The use of recall in making recognition decisions:*

The use of recall in recognition studies has been studied and modeled for many years (Mandler, Pearlstone & Koopmans, 1969; Juola, Fischler, Wood & Atkinson, 1971; see the extended discussion in Gillund & Shiffrin, 1984). In recent years, a design used to address this issue required participants to judge whether they ‘remember’ (i.e. remember details and context of storage) or ‘know’ (i.e. have a sense of familiarity) an item judged old (e.g., Gardiner & Java, 1990). Gillund and Shiffrin (1984) discussed the need for a process of recall-to-reject when foils are highly similar to targets, and cited relevant research. Shiffrin and Steyvers (1997) embedded differentiation in bREM, based on earlier research by Ratcliff, Clark, and Shiffrin (1990); differentiation required strengthening traces recalled during list study; they also discussed the need for recall operating during testing.

This extremely truncated review of results and modeling suggests the roles of several important recognition processes missing from bREM that will be included in the modeling of the present studies. This listing is in no way meant to cover all the important processes that ought to be incorporated in a reasonably complete account of recognition. As an illustration of this point we next mention briefly four processes not incorporated in the extended REM model for the present studies. These are mentioned just to make the point that the present modeling is meant to take a modest step toward a complete recognition model.

A] Associations between items studied. These play an important role in the modeling of TCM (Howard & Kahana, —, and its successor models) where they act as context. Alternatively associations can be treated as content, but this pens up many avenues for modeling and we cannot cover them in this report (see Cox & Shiffrin, 2017; Criss & Cox, —)..

B] Distinctions among content features. Low level features like shape and color play quite a different role than high level features like semantics and meaning, This is again a large field that would open up many avenues for modeling.

C] Dynamics: Different feature types likely arrive in a form that can be used at different points in time (e.g. shape features can be used prior to meaning features). Dynamics played a key role in the modeling of Cox and Shiffrin (2017).

D] Reconstruction of context after significant delay until test. E.g. see Bauml, Meixensperger and Hirsch (in press).

We turn now to the designs of the studies, and the rationale for those designs.

Overview of the designs of the two experiments.

Each study began with ten study-test recognition lists. Each study list contained new pictures not seen earlier in the experiment. Each list was followed by old/new recognition tests, half from the list just studied (targets) and half not from the list just studied (foils). This procedure matches that from most recognition studies.

However, we are presently interested in context change and event memory that extends back in time past the current study list. Thus in the present designs there is a final testing phase after the ten initial lists have been studied and tested. In this final phase, 50% of the test pictures are targets. Targets are defined as any picture seen in some form in the initial ten lists: Such targets included pictures studied but not tested, pictures studied and tested, and pictures tested only as foils. The final phase also included 50% new foils, pictures not seen previously in any form. Presuming that context changes from list to list, final testing allows us to assess such changes. In addition, we wanted to assess the possibility that participants can make use of the storage context during final testing. Thus in Experiment 1, three different orders of final testing were used: *random*, *forward* (pictures seen on the list 1 were targets for the initial set of final tests, then pictures seen on list 2, etc.), and *backward* (pictures seen on list 10 were targets for the initial set of final tests, then pictures seen on list 9, etc.). Participants were informed of the list of origin of the targets before each set of final tests. Because the knowledge of list of origin of the targets helped very little, this manipulation was not used in Experiment 2.

We are also interested in the joint use of familiarity and recall in recognition studies. In the usual paradigms these produce highly correlated responding, because a highly familiar picture is one likely to be recalled, and vice versa. That was the case for the present Experiment 1. The high correlation makes it difficult to assess the processes of familiarity and recall separately. Thus in Experiment 2 three fifths of the foils following each of the initial lists are pictures *not seen on the current study list, but instead seen in some form on the prior list*. Such foils are highly familiar, and discriminating them from current list targets that are also highly familiar likely requires a decision based on more than familiarity. We propose that recall processes are used: In parallel with the computation of familiarity a memory trace is often recalled. The contents of that recalled trace can be evaluated and sometimes the contents contain information about the list of origin of that trace. When that information indicates that the list of origin was the prior list, and not the current list, then a picture with high familiarity can be rejected as a target. In the design of Experiment 2 one source of list-of-origin information is information that the item producing that trace had been tested, because a current list target had not yet been tested. Whatever the source of list-of-origin information, this process of rejecting a highly familiar test item is known as “recall to reject” (e.g. Rotello, Macmillan & Tassel, 2000).

EXPERIMENT 1

METHOD

Participants

Participants were recruited through Prolific Academic (prolific.com), an online research platform. Eligibility requirements included: (1) residence in the United States, (2) minimum age of 18 years, (3) English fluency, and (4) a minimum 98% approval rating on previous Prolific studies to ensure data quality. All procedures were approved by the Institutional Review Board of Indiana University (Protocol #18431). Participants provided informed consent before beginning the experiment. Compensation was provided at an hourly rate of \$10.50, with an estimated completion time of approximately 35 minutes (actual payment of approximately \$6.13).

Stimuli and Apparatus

Stimuli

The experiment used 2,361 unique color photographs of everyday objects and scenes selected from a stimulus set of 2,400 pictures collected by Brady et al. (2008) from Talia Konkle's laboratory. The pictures for each participant were randomly selected from these. Then the pictures for each condition were randomly selected from this subset.

Software and Technical Environment

The experiment was implemented using jsPsych version 7.3.3 (de Leeuw, 2015), a JavaScript library for web-based behavioral experiments providing millisecond-level timing precision. The experiment was hosted on the Cognition.run platform. Technical requirements included desktop computer access and Google Chrome browser usage, which were verified programmatically. Participants were required to use full-screen mode throughout the experiment to minimize distractions. All stimulus images were preloaded before testing to ensure smooth presentation timing.

Design and Procedure

The experiment employed a within-subjects design with two main phases: (1) ten initial lists with study-test cycles and (2) a final recognition test with testing order varied between groups.

Phase 1: Initial Study-Test Cycles

Participants received detailed instructions before beginning the experimental procedure. These instructions explained that the experiment would consist of a practice trial followed by ten formal study-test cycles. For each cycle, participants were told they would complete three sequential stages: first, studying a list of pictures that would appear one at a time on the screen; second, viewing a series of numbers to mentally sum and then typing their answer; and third, completing a recognition test for pictures, pressing the "J" key to indicate they had seen a picture in the just-studied list and pressing the "F" key to indicate they had not seen it in that list. Participants were also told that accumulated accuracy would be displayed on the screen trial by trial as they progressed through the study. The instructions

emphasized that uncertainty about many test pictures was normal and encouraged participants to make their best guess in such cases.

Practice Trial. After providing informed consent and passing browser compatibility checks, participants completed a single practice trial to familiarize themselves with the experimental procedure. This practice trial followed the identical three-stage structure described below but served only as training. Upon completion of the practice trial, participants received brief instructions stating that they had finished the practice and could begin the formal study when ready by pressing the enter key.

Formal Study-Test Cycles

Following the practice trial, participants completed ten formal study-test cycles. Each cycle followed an identical three-stage structure.

Study Phase. After a 1,000 ms fixation cross, 20 unique pictures were presented sequentially. Each picture appeared for 2,000 ms with 100 ms inter-stimulus intervals consisting of blank screens. Participants were instructed to study each picture as it appeared.

Distractor Phase. The distractor phase served to clear short-term memory and create a retention interval between study and test. Following study, participants saw a 3,000 ms screen displaying the instructional prompt "Summing up the digits as they appear," followed by a 1,000 ms fixation cross. Then eight randomly selected digits (ranging from 4 to 9) were presented sequentially, with each digit displayed for 2,000 ms and 1,000 ms gaps between consecutive digits. After all eight digits had been presented, participants saw the prompt "TYPE THE SUM" and entered their calculated sum using the number keys. If participants entered non-numeric characters, a warning message appeared and they were required to re-enter their response. Once a valid numeric response was submitted, participants received immediate feedback displaying either "CORRECT!" or "INCORRECT!" for 2,000 ms.

Recognition Test Phase. Following the distractor task, participants saw a 3,000 ms instructional prompt stating "Now judge if you have seen this picture from the list you JUST studied before the digits," followed by a 1,000 ms fixation cross. Then 20 pictures were presented sequentially for old/new recognition judgments. Of these 20 test pictures, 10 were targets randomly selected from the 20 pictures in the preceding study list, and 10 were foils (novel images that had not been studied). The targets and foils were randomly intermixed. For each picture, participants pressed "J" to indicate "old" (previously studied in the just-completed list) or "F" to indicate "new" (not studied in that list). Participants had up to 3,500 ms to make each response. Following each response, accumulated accuracy for the current recognition test was displayed on the screen (e.g., "Your accumulated accuracy for this trial is: 75%"). Responses faster than 150 ms triggered a "Too fast!" warning message displayed for 1,500 ms. Failures to respond within the 3,500 ms time limit prompted a "You need to respond faster!" message displayed for 1,500 ms.

Item Types Created by the Procedure

This procedure created four distinct item types that would be relevant for the subsequent final recognition test. **Studied-and-Tested (ST)** items were pictures that appeared in the study phase and were subsequently tested as targets in the recognition test. **Studied-Only (SO) items** were pictures that

appeared in the study phase but were not selected as targets for the recognition test. **Test-Only (TO) items** were pictures that appeared only as foils during the initial recognition testing phase and were never studied. **Final-Test-Only (FTO) items** were pictures that appeared exclusively as foils in the final recognition test and had never appeared during any of the ten initial study-test cycles.

Phase 2: Final Recognition Test

Following completion of the tenth study-test cycle, participants transitioned to a final recognition test with modified task instructions. Participants were presented with detailed instructions explaining that they would see a very long series of test pictures. They were explicitly instructed to press the "J" key (old) for any image they had seen at any point during the experiment, whether it had been studied or tested on any list. The instructions specifically emphasized, with highlighted red text, that this included pictures that were not studied but were tested for the first time during the initial recognition tests. Participants were instructed to press the "F" key (new) only for completely novel images that had never appeared before in the experiment.

Participants were further informed that the feedback structure would differ substantially from the initial test phase. Rather than receiving running accuracy feedback after each response, they would instead be shown information about each item's experimental history. For old items (items that had actually appeared in the experiment), they would be told which list number the item had appeared on, with list numbers ranging from 1 for the first list to 10 for the last list. For novel foils (items that had never appeared), they would be informed that the item had never been seen previously.

Test Composition

The final test comprised 420 trials: 210 old items drawn from previous phases and 210 completely novel foils. From each of the 10 study lists, 7 items were randomly sampled from each of the three item types that had actually appeared during the initial phase (ST, SO, and TO), yielding 210 old items total ($10 \text{ lists} \times 3 \text{ item types} \times 7 \text{ items per type} = 210$). These 210 old items were randomly intermixed with 210 novel FTO foils that had never appeared in any prior phase of the experiment.

Between-Subjects Presentation Order Conditions

Participants were randomly assigned to one of three conditions that determined the presentation sequence of the 420 final test trials. The conditions differed in how old items from different initial lists were temporally organized within the final test.

Forward Condition. Participants in the forward condition were informed at the outset that the final test would progress chronologically through the initial lists, beginning with items from List 1 and proceeding sequentially through List 10. The 420 trials were organized into ten blocks of 42 trials each, with each block corresponding to one of the initial study lists in forward temporal order. Before each block of 42 trials, participants received an explicit notification indicating which list number the old items in the upcoming block would be drawn from (e.g., "The items you studied or tested in the upcoming block are from List 3"). Each block of 42 trials contained 21 old items from the designated list (7 ST items, 7

SO items, and 7 TO items) randomly intermixed with 21 novel FTO foils. Within each block of 42 trials, all items were presented in random order.

Backward Condition. Participants in the backward condition were informed that the final test would progress in reverse chronological order, beginning with items from List 10 and proceeding backward sequentially through List 1. The 420 trials were organized into ten blocks of 42 trials each, with each block corresponding to one of the initial study lists in reverse temporal order. Before each block of 42 trials, participants received an explicit notification indicating which list number the old items in the upcoming block would be drawn from (e.g., "The items you studied or tested in the upcoming block are from List 8"). Each block of 42 trials contained 21 old items from the designated list (7 ST items, 7 SO items, and 7 TO items) randomly intermixed with 21 novel FTO foils. Within each block of 42 trials, all items were presented in random order.

Random Condition. Participants in the random condition received all 420 trials in completely random order with no organizational structure by list. These participants were not given any advance information about which list old items were drawn from before any test trial. However, they received the same post-response feedback as participants in the other conditions, with feedback indicating the list number for old items or noting that novel foils had never been seen previously.

Final Test Trial Procedure

Each trial in the final recognition test followed a standard format. Images were presented individually with a response window of up to 4,000 ms. Participants pressed "J" (old) or "F" (new) using the same response mapping as the initial test phase. Responses faster than 150 ms triggered a "Too fast!" warning displayed for 1,500 ms, while failures to respond within the 4,000 ms time limit prompted a "You need to respond faster!" message displayed for 1,500 ms.

Following each response, the test image reappeared for 500 ms with text displayed below indicating its experimental history. For old items, the feedback specified the list on which the item had appeared (e.g., "This item had been studied or tested on list 7"). For novel foils, the feedback stated "This item had never been seen previously." This informative feedback served dual purposes: maintaining participant engagement during the lengthy 420-trial final test and enhancing attention to the temporal structure of the original list presentation order. After the 500 ms feedback display, a 100 ms blank inter-trial interval preceded the next trial.

Dependent measures

Accuracy and response times were recorded for all responses, but only accuracy is reported and used for modeling. **The median response times for every condition are reported in Appendix XXX for archival purposes.** For each of the ten initial lists, recognition accuracy was calculated as the proportion of hits (correct old judgments for target tests) and proportion of correct rejections (correct new responses to foil test). For targets, hits were analyzed by the serial position within the study list. Performance on final tests was also reported by hits and correct rejections, broken down by order of final tests and for targets by order of initial list and serial positions within list. **The figures showing the hits**

and correct rejections, and the tables of hits and correct rejections in Appendix YYY, give for each condition the standard error of the mean, across participants.

The dependent measures paralleled those of Experiment 1, with additional analyses focusing on the confusing foil manipulations:

EXPERIMENT 2

METHOD

Participants

Participant recruitment and eligibility criteria were identical to Experiment 1, with one additional exclusion criterion: participants who had completed Experiment 1 were excluded to prevent contamination from prior exposure to the stimulus set. All procedures were approved by the same Institutional Review Board protocol (Indiana University Protocol #18431), and informed consent procedures matched those of Experiment 1. Compensation was provided at the same hourly rate as Experiment 1 (\$10.50), with an estimated completion time of approximately 45 minutes (actual payment of approximately \$7.88).

Stimuli and Apparatus

Stimuli. Stimuli were drawn from the same database as Experiment 1: unique color photographs from the Brady et al. (2008) stimulus set. However, Experiment 2 required more unique images due to the systematic manipulation of confusing foils. For Experiment 2, exactly 369 unique pictures were required for the initial study-test phase: 45 pictures for List 1 and 324 pictures for Lists 2-10 (36 new items per list). List 1 required more new pictures than subsequent lists because it lacked confusing foils carried forward from previous lists. An additional 246 completely novel images served as final test foils, yielding 615 total unique images.

Software and Technical Environment. The experimental implementation used the same jsPsych version 7.3.3 framework as Experiment 1, with identical technical requirements (desktop computer access, Google Chrome browser, full-screen mode). All stimulus images were preloaded before testing to ensure smooth presentation timing, following the same protocol as Experiment 1. All experimental timing parameters were identical to Experiment 1. Study images were presented for 2,000 ms with 100 ms inter-stimulus intervals, preceded by 1,000 ms fixation crosses. Distractor phases used the same timing (digits presented for 2,000 ms each with 1,000 ms gaps, followed by 2,000 ms feedback displays). Recognition test response windows matched Experiment 1 (3,500 ms for initial tests, 4,000 ms for final test), with identical quality control measures (150 ms response time warnings displayed for 1,500 ms, and non-response prompts). Final test trials included 500 ms feedback following each response, as in Experiment 1.

Design and Procedure

The experiment employed the same within-subjects design structure as Experiment 1: 10 initial study-test cycles followed by a final recognition test. The critical difference was the systematic

manipulation of foil composition during initial test phases. Starting with List 2, three-fifths of the foils on each test list had been seen on the prior list, creating confusing foils that participants were instructed to reject as "new" despite having encountered them previously.

Experimental Design Structure

Stimuli were organized into units of exactly three pictures each. This organization addressed two essential requirements: ensuring adequate statistical power (minimum 3 items per condition per list, yielding 30 data points per condition across all lists per participant) and facilitating counterbalancing of specific images across experimental roles between participants.

The base item types for any given list n were defined as follows: SO (Studied-Only) items were studied but not tested as targets in the initial lists. TO (Test-Only) items appeared only as foils during testing in the initial lists. ST (Studied-and-Tested) items were both studied and tested as targets in the initial lists. FTO (Final-Test-Only) items appeared only as foils in the final test and were never studied in the initial lists.

In addition to these base categories, temporal markers were used to track item carryover between lists. Items appearing only in their current list were denoted as SO, TO, or ST. Items from list n that were reassigned to serve as confusing foils in list $n+1$ were denoted **SO(n)**, **TO(n)**, or **ST(n)**. Conversely, confusing foils in the current list that originated from the preceding list ($n-1$) were denoted **SO($n-1$)**, **TO($n-1$)**, or **ST($n-1$)**. The key constraint was that items could only be carried forward once to serve as confusing foils in the immediately subsequent list, preventing cumulative exposure effects across multiple lists.



Figure 1. (1) Each row represents one list, with the study phase on the left and the recognition test on the right. Each colored box corresponds to a unit of three pictures of the same type, so that every list contains 10 units (30 pictures) at study and 10 units (30

pictures) at test. The different colors indicate assignment type: green boxes are studied-only (SO), blue boxes are studied and later tested (S&T, targets), orange boxes are new foils (TO), and purple boxes are confusing foils—items studied on the previous list that reappear as foils on the current list. Boxes with subscripts (e.g., SO_1 , ST_2) denote repeated items, while non-subscripted boxes indicate items that are unique to that list. (2) **List 1 differs from subsequent lists** in that it contains no confusing foils, because there is no preceding list. Instead, the “slots” that would otherwise be filled by repeated items are filled by additional TO foils. From **List 2 onward, the structure stabilizes**: five studied-only units, five studied-and-tested units, and in the test phase four new target units (blue), one subscripted target unit (blue), two new foil units (orange), and three confusing foil units (purple) carried over from the immediately preceding list. The dashed arrows from List 1 to List 2 illustrate this carry-over: items studied in List 1 (e.g., SO_1 , ST_1 , TO_1) are recycled as confusing foils in the test phase of List 2. Similarly, the arrows from List 2 to List 3 show the next cycle of carry-over (e.g., SO_2 , ST_2 , TO_2 appearing as confusing foils in List 3). List n generalizes this process, where $SO(n-1)$, $ST(n-1)$, and $TO(n-1)$, are carried forward. This design ensures that List 1 introduces **15 unique picture units**, while each subsequent list introduces **12 new units** (because the three purple confusing foils are repeats rather than new pictures). Across lists, the non-subscripted labels (SO, TO, ST) stand for categories, but the actual pictures are unique exemplars within each list.

Each list contained 10 units at study (30 pictures total) and 10 units at test (30 pictures total). List 1 differed from subsequent lists in that it contained no confusing foils because there was no preceding list. Instead, the positions that would otherwise be filled by repeated items were filled by additional TO foils. From List 2 onward, the structure stabilized: five studied-only units, five studied-and-tested units, and in the test phase four new target units, one repeated target unit carried over from the previous list, two new foil units, and three confusing foil units (one each of $SO(n-1)$, $TO(n-1)$, and $ST(n-1)$) carried over from the immediately preceding list. This design ensured that List 1 introduced 15 unique picture units, while each subsequent list introduced 12 new units (because the three confusing foils were repeats rather than new pictures).

Phase 1: Initial Study-Test Cycles

Participants received detailed initial instructions explaining the three-stage structure of each cycle. The instructions stated that participants would see a list of 30 pictures appearing one at a time on the screen, followed by a series of numbers to add mentally before typing the sum, and finally a series of test pictures requiring old/new recognition judgments. Participants were told to press the "J" key if they recognized a picture as one they had seen in the current list and to press the "F" key if they did not recognize it as being from the current list. The instructions emphasized that accumulated accuracy would be displayed throughout the experiment and that participants should try to achieve average performance higher than 60%, with chance level being 50%. The instructions also noted that participants would begin with a practice list followed by nine formal test lists, and importantly, that uncertainty about some test pictures was normal but they should make their best guess.

Practice Trial and Transition to Formal Lists

After providing informed consent and passing browser compatibility checks, participants completed List 1 as a practice trial. This initial list served to familiarize participants with the experimental procedure while also introducing the first set of items that would later serve as confusing foils in List 2.

Prior to List 2, participants received instructions informing them that some test items in subsequent lists might include images they had encountered in previous lists, and that they should respond 'new' to such items since they had not been studied in the current list.

Study-Test Cycle Structure

The ten study-test cycles followed the identical three-stage structure as Experiment 1: study phase, distractor task, and recognition phase.

Study Phase. Procedures were identical to Experiment 1, except that 30 pictures (rather than 20) were presented sequentially. Each picture appeared for 2,000 ms with 100 ms inter-stimulus intervals, preceded by a 1,000 ms fixation cross. For List 1, all 30 studied pictures were new images that had never appeared previously in the experiment. For Lists 2-10, all 30 studied pictures were also new images that had never been studied before, though some of these pictures might later appear as confusing foils in the subsequent list.

Distractor Phase. The arithmetic task was identical to Experiment 1, involving the presentation and summing of eight randomly selected digits to prevent rehearsal between study and test phases. The timing and feedback procedures matched those of Experiment 1 exactly.

Recognition Phase. Following the distractor task, participants saw a 3,000 ms instructional prompt stating "Now judge if you have seen this picture from the list you JUST studied before the digits," followed by a 1,000 ms fixation cross. Then 30 pictures were presented sequentially for old/new recognition judgments. Recognition test feedback was identical to Experiment 1, with running accuracy displayed and updated immediately after each response within the current list (e.g., "Your accumulated accuracy for this list is: 75%"). Response timing, warnings, and quality control measures matched Experiment 1.

For List 1, the 30 test pictures consisted of targets randomly selected from the 30 studied items and novel foils, using the same response mapping as Experiment 1 ("J" for old, "F" for new). For Lists 2-10, each test contained targets from studied items, novel foils, and critically, confusing foils representing different exposure histories from the previous list. Specifically, each test in Lists 2-10 included confusing foils of type SO(n-1), TO(n-1), and ST(n-1), where n-1 refers to the immediately preceding list. Participants were expected to correctly reject these confusing foils as "new" despite having encountered them on the previous list, because they had not been studied in the current list.

Phase 2: Final Recognition Test

Final test procedures closely paralleled Experiment 1, with modifications to accommodate the confusing foil manipulation. Participants received instructions identical in structure to Experiment 1: they were told they had completed the first part of the study and would now see a very long series of test pictures. They were explicitly instructed to press the "J" key (old) for any image encountered at any point during the initial study-test phase, whether it had been studied or tested on any list, and to press the "F" key (new) only for completely novel images. The instructions emphasized that accumulated accuracy would be presented after each response, as it had been during the initial test phase.

Test Composition

The final test comprised 492 trials: 246 items drawn from previous phases and 246 completely novel foils. The 246 old items were sampled at a two-thirds ratio from the 369 items that had appeared during the initial test phase, following the same balanced structure as Experiment 1 to ensure adequate representation of all item types and exposure histories.

Test Order

All participants in Experiment 2 received items in a single randomized presentation order, with all previously encountered and novel items thoroughly intermixed throughout the test sequence. Unlike Experiment 1, which included forward, backward, and random presentation conditions, Experiment 2 employed only the random condition. This decision was made due to the many new conditions created by the confusing foil manipulation and the need to obtain reliable data for each condition. Future research would be valuable to examine confusing foils as in Experiment 2 while also implementing the test order manipulations of Experiment 1.

Final Test Procedure

Each trial followed the same format as Experiment 1. Images were presented individually with a response window of up to 4,000 ms. Participants pressed "J" (old) or "F" (new) using the same response mapping as the initial test phase. Responses faster than 150 ms triggered warnings, and non-responses prompted speed warnings, with timing identical to Experiment 1.

Following each response, participants received feedback adapted from the random order condition of Experiment 1. For items that had appeared during the initial phase, the feedback indicated the item's specific experimental history. Because confusing foils appeared on multiple lists, the feedback for such items specified both list appearances (e.g., "This item had been studied or tested on list 3 and list 4"). For items that appeared on only a single list, the feedback specified that list (e.g., "This item had been studied or tested on list 7"). For novel foils, the feedback stated "This item had never been seen previously." This informative feedback maintained engagement during the lengthy 492-trial final test while providing participants with information about the temporal structure of item presentations.

Dependent Measures

Recognition accuracy and response times were recorded for each study-test cycle, with particular focus on false alarm rates for different confusing foil types (SO(n-1), TO(n-1), ST(n-1)). Overall recognition accuracy in the final test was analyzed by item type and original exposure history, extending Experiment 1's analyses to examine how different prior exposure types influenced long-term recognition performance. The dependent measures paralleled those of Experiment 1, with additional analyses focusing on the confusing foil manipulations and their effects on both initial and final test performance.

Results of Experiment 1

206 participants completed the experiment through online recruitment. Participants were randomly assigned to one of three experimental conditions: backward ($n = 50$), forward ($n = 53$), or random ($n = 103$).

Data quality screening was conducted using multiple exclusion criteria following standard practices in cognitive psychology research (Van Selst & Jolicoeur, 1994). At the participant level, individuals were excluded if they met any of the following criteria: (a) overall task accuracy more than 2.5 standard deviations below the sample mean, (b) initial test performance more than 2.5 standard deviations below the sample mean, (c) median response time more than 3 standard deviations from the sample mean, (d) more than 10% of responses faster than 200 ms (indicating insufficient processing time), or (e) performance decline exceeding 30% from initial to final test (indicating disengagement or fatigue). At the trial level, responses faster than 150 ms or slower than 3,500 ms were excluded from all analyses, consistent with the experiment's response window parameters.

Ten participants (4.9%) were excluded based on participant-level criteria. Specifically, nine participants exhibited extremely low accuracy ($M = 0.54$, $SD = 0.03$, range = 0.51–0.59) combined with additional indicators of insufficient task engagement, including atypical response patterns ($z > 3.0$), rushed responding ($>10\%$ responses < 200 ms), or severe performance decline ($>30\%$ decrease). One additional participant was excluded due to extreme initial test performance ($z = -5.16$) followed by anomalous learning gains ($z = 6.89$), suggesting initial inattention rather than genuine inability. Participant exclusions were distributed across conditions: backward ($n = 2$, 4.0%), forward ($n = 5$, 9.4%), and random ($n = 3$, 2.9%). At the trial level, 351 trials (0.28% of all test trials) were excluded due to response times outside the 150–3,500 ms window (212 trials < 150 ms, 139 trials $> 3,500$ ms).

The final sample consisted of 196 participants (backward: $n = 48$; forward: $n = 48$; random: $n = 100$) contributing 126,876 valid test trials. All participants provided informed consent and were compensated for their time in accordance with institutional review board guidelines.

All model simulations were performed through simulation-based estimation implemented in the Julia programming language (Version 1.10.9; Bezanson et al., 2017). The main results will be displayed in figures as well as given as values in the appendices. In order that the figures showing the data not be repeated later, the predictions of the augmented REM model are shown along with the results. **The data are shown along with standard errors of the mean (across participants).**

Recognition accuracy was analyzed using generalized linear mixed models (GLMM) with binomial error distribution and logit link function. Seven separate analyses examined position effects across different experimental phases: Initial Test Phase: (1) study position effects, (2) test position effects, and (3) between-list effects. Final Test Phase: (4) within-study position effects, (5) within-test position effects, (6) between-session effects by final order, and (7) between-session effects by initial order. All the statistical models included position variables (linear and quadratic terms), item type, and their interactions as fixed effects. Random effects included participant intercepts and position slopes where appropriate. Position variables were orthogonal polynomials to reduce multicollinearity. Models were fitted using maximum likelihood estimation in R using the lme4 package (citation). Significance was evaluated at $\alpha = .05$. Item-type-specific trends were extracted using estimated marginal means with the emmeans package.

Initial List Study and Test Performance

Within-List Effects During Initial Testing

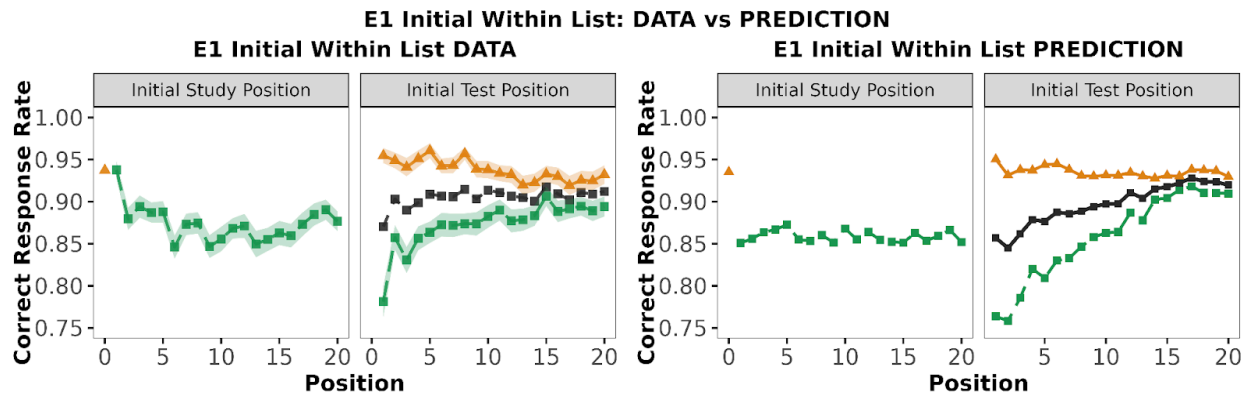


Fig 2.1.a.b (data); 2.2.a.b (predicted): Experiment 1 Initial lists study and test position effects within list. Panel a: study position; panel b: test position. **blue/green = targets, hits (H); orange = foils correct rejections (CR); black = average performance ((H+CR)/2).**

Study Position Effects:

Study position often shows serial position effects, as did the present study, as depicted in **Figure 2.1.a**, left panel. Such effects have been modeled many times (e.g. Atkinson & Shiffrin, 1968). [These serial position effects are not a primary interest of the present investigations and will not be modeled. Instead the modeling will predict performance averaged across study position].

The study position analysis was limited to target items, as foil items do not have study positions. Target items showed a significant negative linear trend, $b = -41.57$, $SE = 2.32$, $z = -17.94$, $p < .001$, 95% CI $[-46.11, -37.02]$, indicating declining accuracy across study positions. However, this was accompanied by a strong positive quadratic effect, $b = 38.21$, $SE = 2.31$, $z = 16.56$, $p < .001$, 95% CI $[33.69, 42.73]$, indicating a pronounced U-shaped serial position curve with better memory for items studied at the beginning and end of each list compared to the middle positions. Target items showed an estimated marginal mean accuracy of 2.56 on the logit scale ($SE = 0.078$), corresponding to approximately 92.8% accuracy. The overall intercept was highly significant, $b = 2.85$, $SE = 0.082$, $z = 34.89$, $p < .001$, 95% CI $[2.69, 3.01]$. Item type showed a significant main effect, with target items performing somewhat lower than the baseline, $b = -0.29$, $SE = 0.058$, $z = -4.91$, $p < .001$, 95% CI $[-0.40, -0.17]$.

Test Position Effects:

Test position effects are important results that help us augment the bREM model. These test position effects are shown in **Figure 2.1.b**. CRs were considerably higher than hits (as is often found, and

as is predicted by bREM). Hs increased strongly over tests. CRs decreased less strongly over tests. Overall performance increased slightly over tests. These results are important because they show the opposite of output interference (see later discussion).

Test position effects differed dramatically by item type, with significant linear and quadratic interactions, as shown in Figure 2.1.b. The baseline linear trend for foil items showed a strong decline, $b = -37.10$, $SE = 3.52$, $z = -10.55$, $p < .001$, 95% CI $[-43.99, -30.21]$, indicating that correct rejections decreased across test positions. Foil items also showed a modest positive quadratic effect, $b = 9.34$, $SE = 3.05$, $z = 3.06$, $p = .002$, 95% CI $[3.36, 15.32]$, indicating a U-shaped recovery pattern with somewhat better performance at early and late test positions.

In stark contrast, target items showed a strong positive linear trend interaction, $b = 74.95$, $SE = 3.62$, $z = 20.68$, $p < .001$, 95% CI $[67.85, 82.05]$, yielding an estimated marginal linear trend of $b = 37.85$ ($SE = 3.57$, $z = 10.60$, $p < .001$, 95% CI $[30.86, 44.84]$), indicating that hits improved dramatically across test positions. Target items also showed a significant negative quadratic interaction, $b = -25.94$, $SE = 3.85$, $z = -6.74$, $p < .001$, 95% CI $[-33.47, -18.40]$, yielding an estimated marginal quadratic trend of $b = -16.59$ ($SE = 3.47$, $z = -4.78$, $p < .001$, 95% CI $[-23.39, -9.80]$), indicating an inverted U-shaped pattern with particularly strong performance gains that plateaued toward the end of testing.

Overall item type differences were substantial. Foil items showed significantly higher overall accuracy (estimated marginal mean = 3.07 on logit scale, $SE = 0.079$, corresponding to approximately 95.6%) compared to target items (EMM = 2.24, $SE = 0.075$, corresponding to approximately 90.4%), $b = -0.83$, $SE = 0.039$, $z = -21.25$, $p < .001$, 95% CI $[-0.91, -0.75]$. Pairwise comparisons with Tukey correction confirmed this significant difference ($p < .001$).

Between-List Effects During Initial Testing

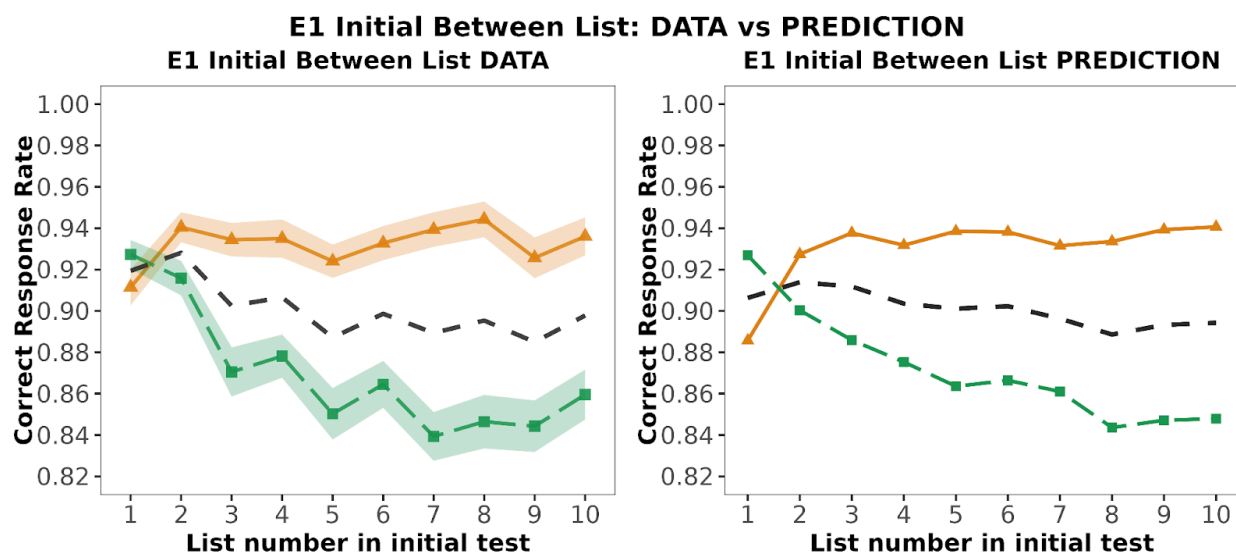


Figure 3.a (data); 3.b (predicted): Experiment 1 between list performance. Green: targets Hs; orange: foil CRs; black: average performance. **Error ribbons** represent ± 1 standard error of the mean.

Across lists we see clear evidence of output interference, as shown in Figure 3.a. Overall performance drops due to a strong decrease in Hs (green). CRs (orange) are largely stable over lists.

The baseline linear trend for foil items showed no significant effect, $b = -4.21$, $SE = 3.59$, $z = -1.17$, $p = .242$, 95% CI [-11.25, 2.83], indicating that correct rejections remained relatively stable across lists. However, foil items showed a significant positive quadratic effect, $b = 10.36$, $SE = 2.82$, $z = 3.67$, $p < .001$, 95% CI [4.83, 15.89], indicating a U-shaped pattern with some recovery in later lists.

In contrast, target items showed dramatically different patterns. The linear trend interaction for targets was strongly negative, $b = -52.10$, $SE = 3.38$, $z = -15.41$, $p < .001$, 95% CI [-58.72, -45.47], yielding an estimated marginal linear trend of $b = -56.30$ ($SE = 4.47$, $z = -12.59$, $p < .001$, 95% CI [-65.07, -47.53]), indicating substantial interference effects with declining hits across lists. Target items also showed a strong positive quadratic interaction, $b = 32.43$, $SE = 3.53$, $z = 9.19$, $p < .001$, 95% CI [25.51, 39.35], yielding an estimated marginal quadratic trend of $b = 42.79$ ($SE = 3.27$, $z = 13.09$, $p < .001$, 95% CI [36.38, 49.19]), indicating strong U-shaped recovery in later lists as participants adapted to the task.

Overall item type differences persisted across lists. Foil items showed significantly higher overall accuracy (estimated marginal mean = 3.08 on logit scale, $SE = 0.079$, corresponding to approximately 95.6%) compared to target items (EMM = 2.29, $SE = 0.076$, corresponding to approximately 90.8%), $b = -0.79$, $SE = 0.039$, $z = -20.37$, $p < .001$, 95% CI [-0.87, -0.72]. Pairwise comparisons with Tukey correction confirmed this significant difference ($p < .001$). Averaging across both item types, there was a substantial linear decline across lists (average estimated marginal trend $b = -30.26$) and a strong U-shaped recovery pattern (average estimated marginal quadratic trend $b = 26.58$), indicating significant output interference effects with recovery that applied across both item types combined, though more pronounced for targets.

Final Recognition Test Performance

Within-List Final Test Results

Study Position Effects:

Figure 4.1.a.b: shows that study position shows little effect of initial study position. This result is notably different from the serial study position effects observed during study of the initial lists. This result suggests that the initial list study position effects were not due to changes in trace strength; forgetting did take place between initial lists and final testing, but it is hard to imagine forgetting that would only affect the primacy and recency portions of the study lists.

The analysis revealed no significant linear trends for either SO or ST items. SO items showed a non-significant declining trend (estimated marginal $b = -5.12$, $SE = 2.82$, $z = -1.81$, $p = .070$, 95% CI [-10.65, 0.41]), while ST items showed a similar non-significant pattern (estimated marginal $b = -6.93$, $SE = 3.76$, $z = -1.83$, $p = .154$, 95% CI [-16.44, 2.58]).

Item-type-specific effects were substantial. ST items ($M = 92.8\%$, estimated marginal mean = 2.28 on logit scale, $SE = 0.061$) substantially outperformed SO items ($M = 68.9\%$, EMM = 0.80, $SE =$

0.057), with the difference highly significant, $b = 1.48$, $SE = 0.035$, $z = 42.06$, $p < .001$, 95% CI [1.41, 1.55]. Pairwise comparisons with Tukey correction confirmed this large and significant difference ($p < .001$), indicating that items both studied and tested during initial learning were remembered far better than items only studied.

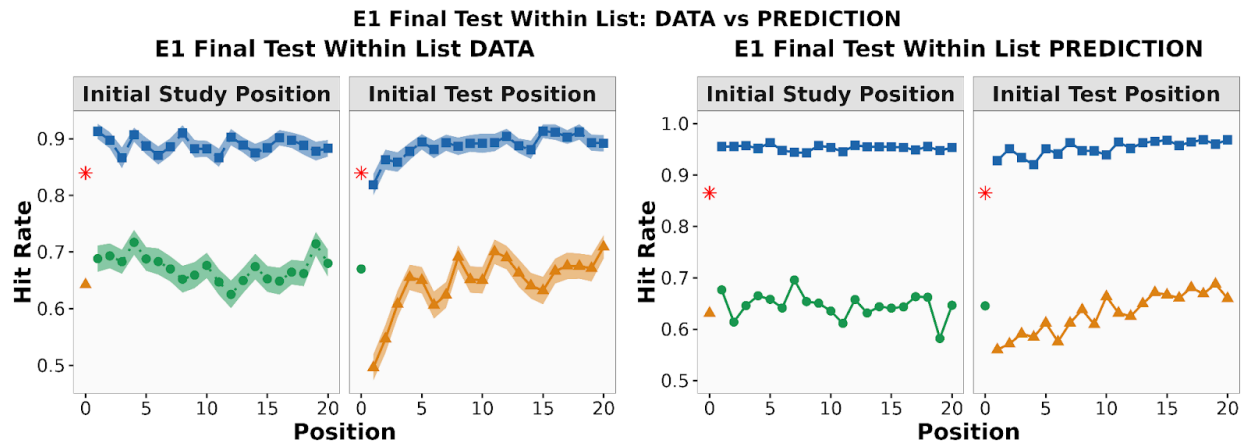


Figure 4.1.a.b (data); 4.2.a.b (predicted). Final test within-list study position effects (4.1.a) and final test within-list test position effects (4.1.b). (blue = target Hs; orange = test-only (foil)Hs, green = studied-only Hs), with the addition of red stars = final foil CRs; **X = foil final test CRs**. Error ribbons represent ± 1 standard error of the mean. **(LEA: CHANGE RED TO A LARGE X)**

Test Position Effects:

Final test accuracy strongly increased with initial list test position, suggesting initial list test position effects were due to trace strength differences. The increase in foils was opposite the initial list decrease, suggesting the initial list decrease may have been due to a temporary shift in decision criterion.

ST items showed a significant linear improvement (estimated marginal $b = 22.55$, $SE = 2.87$, $z = 6.47$, $p < .001$, 95% CI [15.72, 29.38]), indicating that items tested later in the initial lists showed better final test performance. TO items showed an even stronger linear improvement (estimated marginal $b = 27.03$, $SE = 2.68$, $z = 8.78$, $p < .001$, 95% CI [21.00, 33.06]), with the increase in accuracy for foils being opposite to the initial list decline, suggesting that the initial list decrease may have been due to a temporary shift in decision criterion while the final test increase reflects true memory strength differences.

Item type differences revealed a clear performance hierarchy. ST items showed substantially higher accuracy ($M = 90.4\%$, estimated marginal mean = 2.24 on logit scale, $SE = 0.054$) compared to TO items ($M = 64.9\%$, $EMM = 0.64$, $SE = 0.049$), $b = -1.60$, $SE = 0.035$, $z = -45.59$, $p < .001$, 95% CI [-1.67, -1.53]. Pairwise comparisons with Tukey correction confirmed this large and significant difference ($p < .001$), demonstrating the substantial benefit of testing during initial learning.

Between-List Final Test Results

Note that the backward and forward results are mirrors of each other measured by input list position or final test position. Thus in Figure 5, the first and second columns of row 1 show the same data reversed, and the first and second columns of row 2 show the same data.

Final test-ing as a function of initial list position~~order~~:

Figure 5.1.a shows output interference (OI) for all picture types. The final test order manipulation (forward, backward, random) ~~had~~ showed effects important for modeling. ~~a small effect, at, once OI is taken into account.~~ A] Initial targets have the highest final Hs, suggesting they were stored strongly; there is little OI for these pictures, possibly due to ceiling effects.. B] Pictures studied only and pictures tested only had much lower Hs and differed little from each other suggesting they had been stored much more weakly than targets (i.e studied and tested pictures) but were stored in similar fashion: to each other. C] Backward showed strong recency, and strong OI compared to the other conditions. It is assumed that this is due to two factors acting in concert: Use of changing context in final test retrieval that produces recency – better performance for the most recent lists seen – and output interference (OI) that causes performance to decrease as testing continues (see the modeling section for a discussion of the factors causing OI). D] Backward also showing some primacy. The best account is a gradual reconstruction of changing context as the testing moves to earlier lists. This idea will be used to explain the forward data. E] The forward results show greatly reduced performance as lists continue. Our best account is reconstruction of changing context. E.g. When the participants are told that the targets are coming from the first list, they manage to reconstruct context that is more similar to that stored originally than the current context. This better context continues to be used as testing continues. F] The random condition provides feedback that tells the participant what list a target came from, but that list randomly jumps among lists as testing continues. Would this feedback nonetheless lead to some reconstruction of changing context? This is unclear, and the modeling assumes the changing context used is that used at the outset of the backward condition. ???

We fit a model that included condition (forward, random, backward) as a main effect and tested two-way interactions between position effects and item type. Foil items (completely novel items on the final test) showed a substantial baseline decline (estimated marginal $b = -32.7$, $SE = 3.96$, $z = -8.3$, $p < .001$, 95% CI [-40.5, -25.0]). SO items showed a similar decline (estimated marginal $b = -36.8$, $SE = 5.40$, $z = -6.8$, $p < .001$, 95% CI [-47.4, -26.2]). ST items showed the steepest decline (estimated marginal $b = -86.2$, $SE = 8.82$, $z = -9.8$, $p < .001$, 95% CI [-103.5, -69.0]), indicating that the most highly studied items suffered the greatest output interference. TO items showed an intermediate decline (estimated marginal $b = -60.9$, $SE = 5.35$, $z = -11.4$, $p < .001$, 95% CI [-71.4, -50.5]).

Each item type also exhibited significant quadratic recovery, yielding U-shaped performance curves with higher accuracy at the beginning and end of the test. Foil items showed modest recovery (estimated marginal $b = 7.82$, $SE = 3.94$, $z = 2.0$, $p = .046$, 95% CI [0.10, 15.5]). SO items showed moderate recovery (estimated marginal $b = 28.64$, $SE = 5.36$, $z = 5.3$, $p < .001$, 95% CI [18.13, 39.1]). ST items showed the strongest recovery (estimated marginal $b = 52.01$, $SE = 8.45$, $z = 6.2$, $p < .001$, 95% CI [35.45, 68.6]), and TO items also showed substantial recovery (estimated marginal $b = 31.29$, $SE = 5.33$, $z = 5.9$, $p < .001$, 95% CI [20.84, 41.7]).



Figure 5.1.a.b (data); 5.2.a.b (predicted). All results are conditioned on correct initial list responses. a: Final test order (combined into ten groups). b: Initial list order. The three rows of panels show the backward, forward, and random final test order conditions, as labeled. In row one, the second column shows the same results as the first, but reversed; in row two the second column shows the same results as the first. **(blue squares** = final test Hs for pictures studied and tested in the initial lists (**targets, ST**); **green circles** = final test Hs for pictures studied but not tested initially (**SO**); **orange triangles** = Final test Hs for pictures tested initially as foils (**TO**); **Red Asterisks** = final test CRs for foils (pictures not seen in the initial lists; **CRs**), **Error ribbons** represent ± 1 standard error of the mean.

We fit a model that included condition (forward, random, backward) as a main effect and tested two-way interactions between position effects and item type. Foil items (completely novel items on the final test) showed a substantial baseline decline (estimated marginal $b = -32.7$, $SE = 3.96$, $z = -8.3$, $p < .001$, 95% CI [-40.5, -25.0]). SO items showed a similar decline (estimated marginal $b = -36.8$, $SE = 5.40$, $z = -6.8$, $p < .001$, 95% CI [-47.4, -26.2]). ST items showed the steepest decline (estimated marginal $b = -86.2$, $SE = 8.82$, $z = -9.8$, $p < .001$, 95% CI [-103.5, -69.0]), indicating that the most highly studied items suffered the greatest output interference. TO items showed an intermediate decline (estimated marginal $b = -60.9$, $SE = 5.35$, $z = -11.4$, $p < .001$, 95% CI [-71.4, -50.5]).

Each item type also exhibited significant quadratic recovery, yielding U-shaped performance curves with higher accuracy at the beginning and end of the test. Foil items showed modest recovery

(estimated marginal $b = -7.82$, $SE = 3.94$, $z = 2.0$, $p = .046$, 95% CI [0.10, 15.5]). SO items showed moderate recovery (estimated marginal $b = 28.64$, $SE = 5.36$, $z = 5.3$, $p < .001$, 95% CI [18.13, 39.1]). ST items showed the strongest recovery (estimated marginal $b = 52.01$, $SE = 8.45$, $z = 6.2$, $p < .001$, 95% CI [35.45, 68.6]), and TO items also showed substantial recovery (estimated marginal $b = 31.29$, $SE = 5.33$, $z = 5.9$, $p < .001$, 95% CI [20.84, 41.7]).

Averaged across final test positions, planned pairwise comparisons with Tukey correction revealed a clear performance hierarchy (all pairwise differences $p < .001$): ST items showed the highest performance ($M = 90.2\%$, $EMM = 2.22$, $SE = 0.045$), followed by final test foils ($M = 85.0\%$, $EMM = 1.73$, $SE = 0.037$), SO items ($M = 68.0\%$, $EMM = 0.75$, $SE = 0.039$), and TO items showing the lowest performance ($M = 65.2\%$, $EMM = 0.63$, $SE = 0.039$). Specific pairwise contrasts confirmed this hierarchy: ST vs. foil ($b = 0.49$, $SE = 0.033$, $z = 14.71$, $p < .001$), ST vs. SO ($b = 1.46$, $SE = 0.036$, $z = 41.22$, $p < .001$), ST vs. TO ($b = 1.59$, $SE = 0.035$, $z = 44.85$, $p < .001$), foil vs. SO ($b = 0.98$, $SE = 0.024$, $z = 40.95$, $p < .001$), foil vs. TO ($b = 1.10$, $SE = 0.024$, $z = 46.48$, $p < .001$), and SO vs. TO ($b = 0.12$, $SE = 0.027$, $z = 4.56$, $p < .001$).

Initial List Position

Figure 5.1.b shows final test performance as a function of initial list position. The backward and forward conditions show recency: Best performance for the most recent lists seen, regardless of the order of testing. The random condition shows reduced effects of initial list position, likely due to the random order of final testing.

To examine how retention varied as a function of the list on which items were originally encountered, we fit a model examining initial list position effects across all item types (Figure 5.1.b). The baseline linear trend for foil items showed no significant effect (estimated marginal $b = -0.09$, $SE = 1.41$, $z = -0.06$, $p = .949$, 95% CI [-2.85, 2.67]), as these items had no initial list positions. However, all previously encountered item types demonstrated significant linear improvement, indicating recency effects. SO items showed moderate improvement (estimated marginal $b = 14.22$, $SE = 1.88$, $z = 7.6$, $p < .001$, 95% CI [10.54, 17.91]). ST items showed similarly strong improvement (estimated marginal $b = 16.76$, $SE = 1.87$, $z = 9.0$, $p < .001$, 95% CI [13.09, 20.42]). TO items showed the strongest linear improvement (estimated marginal $b = 39.50$, $SE = 1.81$, $z = 21.8$, $p < .001$, 95% CI [35.96, 43.05]), indicating dramatically better retention for items from more recent lists.

Each item type also showed significant quadratic acceleration, indicating accelerating improvement over positions. Foil items showed modest acceleration (estimated marginal $b = 5.59$, $SE = 1.44$, $z = 3.9$, $p < .001$, 95% CI [2.77, 8.42]). SO items showed substantial acceleration (estimated marginal $b = 26.07$, $SE = 1.87$, $z = 13.9$, $p < .001$, 95% CI [22.40, 29.73]). ST items showed the strongest acceleration (estimated marginal $b = 56.32$, $SE = 1.87$, $z = 30.1$, $p < .001$, 95% CI [52.66, 59.99]). TO items also showed strong acceleration (estimated marginal $b = 32.75$, $SE = 2.05$, $z = 16.0$, $p < .001$, 95% CI [28.72, 36.77]).

Averaged across initial list positions, planned pairwise comparisons with Tukey correction revealed a clear performance hierarchy (all pairwise differences $p < .001$): ST items showed the highest performance ($M = 89.9\%$, $EMM = 2.19$, $SE = 0.044$), followed by final test foils ($M = 84.9\%$, $EMM = 1.73$, $SE = 0.037$), SO items ($M = 67.9\%$, $EMM = 0.75$, $SE = 0.039$), and TO items showing the lowest performance ($M = 65.1\%$, $EMM = 0.63$, $SE = 0.039$). Specific pairwise contrasts confirmed this hierarchy: ST vs. foil ($b = 0.46$, $SE = 0.032$, $z = 14.54$, $p < .001$), ST vs. SO ($b = 1.44$, $SE = 0.034$, $z = 41.72$, $p < .001$), ST vs. TO ($b = 1.56$, $SE = 0.034$, $z = 45.57$, $p < .001$), foil vs. SO ($b = 0.97$, $SE = 0.024$, $z = 40.90$, $p < .001$), foil vs. TO ($b = 1.10$, $SE = 0.024$, $z = 46.63$, $p < .001$), and SO vs. TO ($b = 0.13$, $SE = 0.027$, $z = 4.67$, $p < .001$).

E1 Final Test E1_participant_performance

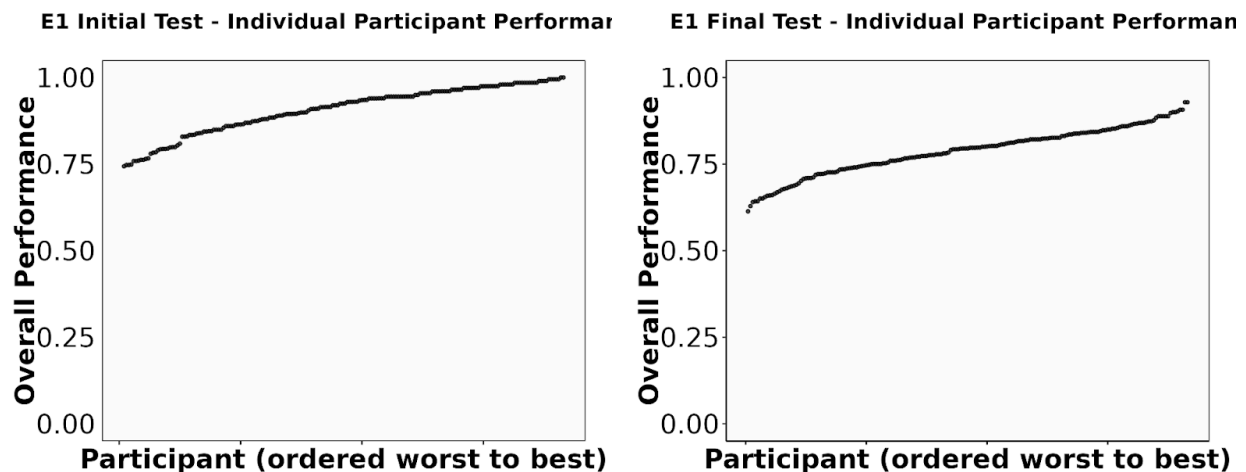


Figure 6.1.2: Experiment 1 individual differences: 6.1: Participant performance on the initial ten lists ordered by participant accuracy; 6.2: Participant performance during final testing ordered by participant accuracy.

Figure 6 shows the very large range of individual differences in Exp. 1. These results are important for modeling: All the data predicted are an average of individual data, and hence a mixture of widely varying results. Under these circumstances, and because the data per condition is too sparse to allow separate modeling of individuals, we cannot expect precise quantitative predictions. Thus the goal of modeling is qualitative: We will produce a model that can predict the trends in the data, and will try to choose parameter values so that the predictions are not too discrepant quantitatively.

Results of Experiment 2

199 participants completed the experiment through online recruitment via Prolific.

Data quality screening was conducted using multiple exclusion criteria following standard practices in cognitive psychology research (Van Selst & Jolicoeur, 1994). At the participant level, individuals were excluded if they met any of the following criteria: (a) below-chance performance (accuracy $< 50\%$), indicating random responding; (b) overall task accuracy more than 2.5 standard deviations below the sample mean; (c) median response time more than 3 standard deviations from the sample mean, indicating atypical response patterns; or (d) more than 10% of responses faster than 200 ms,

indicating insufficient processing time or rushed responding. At the trial level, responses faster than 150 ms or slower than 3,500 ms were excluded from all analyses, consistent with established response window parameters.

13 participants (6.5%) were excluded based on participant-level criteria. Specifically, nine participants exhibited below-chance performance ($M = 0.48$, $SD = 0.01$, range = 0.48–0.50), indicating they were responding randomly or not engaging meaningfully with the task. Two additional participants exhibited extremely low accuracy (22.9% and 29.1%, both $>3 SD$ below the mean) combined with below-chance performance. Three participants were excluded due to extreme response time outliers (median RTs of 281 ms, 1,654 ms, and 1,669 ms; $|z| > 3.3$), with one also exhibiting rushed responding (19.5% of responses < 200 ms). One participant who had been previously identified during initial screening (59.9% accuracy) was also excluded. At the trial level, 479 trials (0.31% of all test trials) were excluded due to response times outside the 150–3,500 ms window (355 trials < 150 ms, 124 trials $> 3,500$ ms).

The final sample consisted of 186 participants contributing 156,214 valid test trials. All participants provided informed consent and were compensated for their time in accordance with institutional review board guidelines.

~~Data from X participants were excluded due to technical issues, failure to maintain full-screen mode, or extremely low performance suggesting disengagement, leaving N participants for analysis. As was done for Experiment 1, figures giving predictions for Experiment 2 are placed adjacent to the figures showing the data, but discussion will be deferred to the modeling section to follow.~~

Recognition accuracy was analyzed using generalized linear mixed-effects models (GLMMs) with a binomial error distribution and logit link, implemented in R with the *lme4* package (Bates, Mächler, Bolker, & Walker, 2015). Seven analyses examined serial position effects across experimental phases: (a) initial test—study position, test position, and between-list effects; and (b) final test—within-study position, within-test position, between-list effects by final order, and between-list effects by initial order. Fixed effects included linear and quadratic position terms, fully interacting with item type¹. Random intercepts for participants were included. Position terms were entered as orthogonal polynomials to reduce multicollinearity, and significance was evaluated at $\alpha = .05$. Item-type-specific trends were estimated with the *emmeans* package (Lenth, 2023).

~~Note: Experiment 2 does not vary test order in the fashion of Experiment 1. Thus the inferences made about the different uses of CC features for different test orders and the differing reconstruction of context for different test orders, is no longer relevant. Testing in Experiment 2 is in random order, and the inferences and model assumptions will be described accordingly.~~

Initial List Study and Test Performance

¹ The maximal models initially specified random intercepts for participants and random slopes for position effects, following recommendations for confirmatory mixed-effects modeling (Barr, Levy, Scheepers, & Tily, 2013). However, several models failed to converge (degenerate Hessians, relative gradients between 0.002–0.019). To achieve stable fits, the random slopes were removed and, when necessary, the quadratic $\times$ item type interaction was dropped. The final convergent specification retained the linear $\times$ item type interaction (unique slopes per item type) and a shared quadratic term.

Within-List Study Position Effects During Initial Testing

Figure 7.1.a shows the existence of serial position effects, as was found in Exp 1. As was the case for Exp 1, these effects were not modeled.

We examined memory performance as a function of study position using a generalized linear mixed-effects model with linear and quadratic position terms. The model included the linear study position term, item type, a shared quadratic study position term, and their interaction, with random intercept.

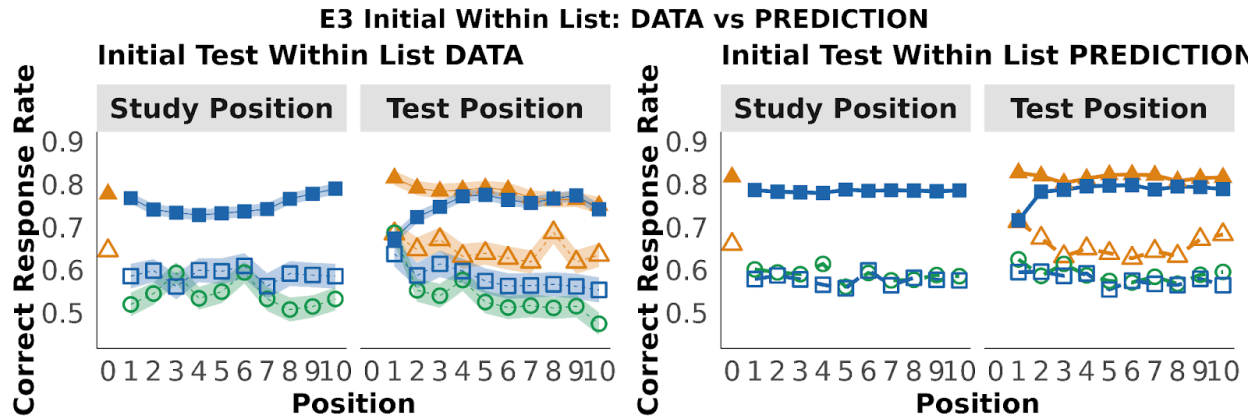


Figure 7.1.a.b and 7.2.a.b: 7.1.a.b; Observed study and test position effects in initial lists, in ten groups, three pictures per group per participant. 7.2.a.b; Model prediction. **blue squares, solid lines:** current list target picture Hs; **orange triangles, solid lines:** current list new foil CRs; **orange triangles, dashed lines:** current list foils that were also foils on the prior trial; **blue squares, dashed lines:** foils on the current list that had been studied and tested on the prior trial. **green circles, dashed lines:** current list foils that had been studied but not tested on the prior trial; **Error ribbons** represent ± 1 standard error of the mean.

The model showed a significant quadratic effect, $b = 27.65$, $SE = 1.67$, $z = 16.55$, $p < .001$, 95% CI [24.38, 30.93], indicating a U-shaped serial position curve across all item types. A significant negative linear component was observed, $b = -6.16$, $SE = 2.18$, $z = -2.82$, $p = .005$, 95% CI [-10.44, -1.89], indicating that the serial position curve showed some asymmetry beyond the dominant U-shaped pattern. Significant main effects of item type emerged: current-list targets, ST ($M = 71.3\%$, logit $b = 0.71$, $SE = 0.04$, $z = 18.78$, $p < .001$), and current-list new foils, TO ($M = 73.6\%$, $b = 0.74$, $SE = 0.04$, $z = 19.46$, $p < .001$), showed higher accuracy than confusing foils. Among confusing foils, SO(n-1) items that had been studied only on the previous list showed the lowest performance ($M = 48.9\%$, $b = -0.49$, $SE = 0.04$, $z = -11.43$, $p < .001$, relative to baseline), while ST(n-1) confusing foils that had been targets on the previous list showed intermediate performance ($M = 57.1\%$, $b = -0.29$, $SE = 0.04$, $z = -6.74$, $p < .001$).

Within-List Test Position Effects During Initial Testing

Figure 7.1.b shows test position effects conditioned on the various types of test items. The solid lines show current list target Hs and current list foil CRs. These show roughly the pattern of Experiment

1, with Hs rising and CRs falling as tests continue. As discussed soon these results do not appear in final testing so are not due to a factor that alters storage strengths as tests continue after each list. Potential explanations will be discussed in the modeling section.

The most interesting results are in the dashed lines that show correct rejections (CRs) for current list foils that had been seen on the prior trial (termed ‘confusing foils’). Due to their presence on the prior trial these test pictures are highly familiar and as a result are given many false alarms (FAs), or equivalently are given low CRs. The most familiar pictures are those studied and tested on the prior trial (as demonstrated by final test results to be shown shortly). If familiarity (REM odds, or some other measure of degree of familiarity) were used to respond, these pictures would be given the greatest false alarms (FAs), and equivalently the lowest CRs. Yet these very familiar pictures produce CRs higher than pictures from the prior trial that were studied only, and not much lower than CRs for foils seen on the prior trial. Such results suggest that participants faced with highly familiar pictures that are not from the current list start using a recall process: They recall a trace likely to be one produced by the test picture and assess the trace contents. When they find information in that trace that identifies the origin of the item producing the trace to be the prior list, they ignore the high familiarity and respond new. This is a process known as ‘recall to reject’ (e.g. Rotello, Macmillan & Tassel, 2000). This process is described in detail in the modeling section.

Test position effects differed dramatically by item type, as shown in Figure 7.1.b. The baseline linear trend (for confusing foils that had been test-only items TO(n-1) on the previous list) showed a strong decline, $b = -10.92$, $SE = 1.90$, $z = -5.75$, $p < .001$, 95% CI $[-14.64, -7.20]$. However, this pattern varied substantially across item types. Current-list targets (ST) showed a strong positive linear trend interaction, $b = 34.45$, $SE = 1.92$, $z = 17.95$, $p < .001$, 95% CI $[30.69, 38.21]$, indicating improving accuracy across test positions (estimated marginal trend $b = 23.53$). Current-list new foils (TO) showed a declining trend (estimated marginal $b = -23.37$, interaction $b = -12.45$, $SE = 2.08$, $z = -5.99$, $p < .001$, 95% CI $[-16.52, -8.38]$). The three types of confusing foils (items from the previous list appearing as foils on the current list) all showed declining accuracy across test positions, but differed in the magnitude of decline. Confusing foils that had been studied-only items on the previous list SO(n-1) showed the steepest decline (estimated marginal $b = -59.34$, interaction $b = -48.42$, $SE = 2.22$, $z = -21.79$, $p < .001$, 95% CI $[-52.77, -44.06]$), followed by the baseline TO(n-1) foils (estimated marginal $b = -10.92$), and confusing foils that had been studied-and-tested targets on the previous list ST(n-1) showed the smallest decline (estimated marginal $b = -17.91$, interaction $b = -6.98$, $SE = 3.23$, $z = -2.17$, $p = .030$, 95% CI $[-13.31, -0.66]$). All item types showed a shared quadratic trend, $b = -11.14$, $SE = 1.81$, $z = -6.16$, $p < .001$, 95% CI $[-14.68, -7.59]$, indicating an inverted U-shaped pattern.

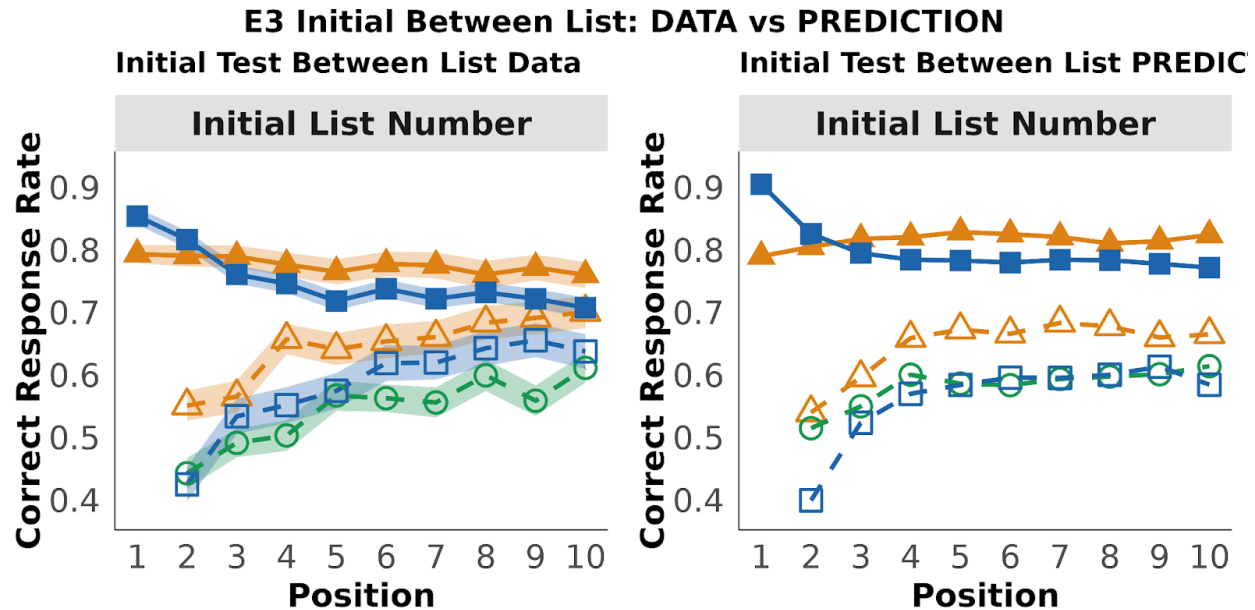


Figure 8.1.2: Correct response rates as a function of list position for different item types. 8.1: Observed data; 8.2: Model predictions. **blue squares** = targets, **orange triangles** = foils, **green circles** = studied-only; **solid shapes** = non-repeated items, **open shapes** = repeated items. **Error ribbons** represent ± 1 standard error of the mean.

Within-List Study Position Effects During Initial Testing

Figure 7.1.a shows the existence of serial position effects, as was found in Exp 1. As was the case for Exp 1, these effects were not modeled.

We examined memory performance as a function of study position using a generalized linear mixed-effects model with linear and quadratic position terms. The model included the linear study position term, item type, a shared quadratic study position term, and their interaction, with random intercepts for participants.

The model showed a significant quadratic effect, $b = 27.65$, $SE = 1.67$, $z = 16.55$, $p < .001$, 95% CI [24.38, 30.93], indicating a U-shaped serial position curve across all item types. A significant negative linear component was observed, $b = -6.16$, $SE = 2.18$, $z = -2.82$, $p = .005$, 95% CI [-10.44, -1.89], indicating that the serial position curve showed some asymmetry beyond the dominant U-shaped pattern. Significant main effects of item type emerged: current-list targets, ST ($M = 71.3\%$, logit $b = 0.71$, $SE = 0.04$, $z = 18.78$, $p < .001$), and current-list new foils, TO ($M = 73.6\%$, $b = 0.74$, $SE = 0.04$, $z = 19.46$, $p < .001$), showed higher accuracy than confusing foils. Among confusing foils, SO($n-1$) items that had been studied only on the previous list showed the lowest performance ($M = 48.9\%$, $b = -0.49$, $SE = 0.04$, $z = -11.43$, $p < .001$, relative to baseline), while ST($n-1$) confusing foils that had been targets on the previous list showed intermediate performance ($M = 57.1\%$, $b = -0.29$, $SE = 0.04$, $z = -6.74$, $p < .001$).

Within-List Test Position Effects During Initial Testing

Figure 7.1.b shows test position effects conditioned on the various types of test items. The solid lines show current list target Hs and current list foil CRs. These show roughly the pattern of Experiment 1, with Hs rising and CRs falling as tests continue. Potential explanations will be discussed in the modeling section. The most interesting results are in the dashed lines that show CRs for current list foils that had been seen on the prior trial (termed ‘confusing foils’). Due to their presence on the prior trial these test pictures are highly familiar and as a result are given many false alarms (FAs), or equivalently are given low CRs. The most familiar pictures are those studied and tested on the prior trial (as demonstrated by final test results to be shown shortly). If familiarity (REM odds) were used to respond, these pictures would be given the greatest false alarms (FAs), and equivalently the lowest CRs. Yet these very familiar pictures produce CRs higher than pictures from the prior trial that were studied only, and not much lower than CRs for foils seen on the prior trial. Such results suggest that participants faced with highly familiar pictures that are not from the current list start using a recall process: They recall a trace likely to be one produced by the test picture and assess the trace contents. When they find information in that trace that identifies the origin of the item producing the trace to be the prior list, they ignore the high familiarity and respond new. This is a process known as ‘recall to reject’ (e.g. Rotello, Macmillan & Tassel, 2000). This process is described in detail in the modeling section.

Test position effects differed dramatically by item type, as shown in Figure 7.1.b. The baseline linear trend (for confusing foils that had been test-only items TO(n-1) on the previous list) showed a strong decline, $b = -10.92$, $SE = 1.90$, $z = -5.75$, $p < .001$, 95% CI [-14.64, -7.20]. However, this pattern varied substantially across item types. Current list targets (ST) showed a strong positive linear trend interaction, $b = 34.45$, $SE = 1.92$, $z = 17.95$, $p < .001$, 95% CI [30.69, 38.21], indicating improving accuracy across test positions (estimated marginal trend $b = 23.53$). Current list new foils (TO) showed a declining trend (estimated marginal $b = -23.37$, interaction $b = -12.45$, $SE = 2.08$, $z = -5.99$, $p < .001$, 95% CI [-16.52, -8.38]). The three types of confusing foils (items from the previous list appearing as foils on the current list) all showed declining accuracy across test positions, but differed in the magnitude of decline. Confusing foils that had been studied-only items on the previous list SO(n-1) showed the steepest decline (estimated marginal $b = -59.34$, interaction $b = -48.42$, $SE = 2.22$, $z = -21.79$, $p < .001$, 95% CI [-52.77, -44.06]), followed by the baseline TO(n-1) foils (estimated marginal $b = -10.92$), and confusing foils that had been studied and tested targets on the previous list ST(n-1) showed the smallest decline (estimated marginal $b = -17.91$, interaction $b = -6.98$, $SE = 3.23$, $z = -2.17$, $p = .030$, 95% CI [-13.31, -0.66]). All item types showed a shared quadratic trend, $b = -11.14$, $SE = 1.81$, $z = -6.16$, $p < .001$, 95% CI [-14.68, -7.59], indicating an inverted U-shaped pattern.

Between List Effects During Initial Testing

Figure 8 shows between list results during initial lists, again conditioned on type of test picture. The relatively constant CRs for current list foils suggests that the decision threshold based on familiarity is changing little if at all. It could be that recall to reject is sometimes used when foils are familiar enough to produce a false alarm (FA), and thereby alter a response from old to new, increasing the level of CRs for all lists. That will be seen when the model is used.

Each of the three types of confusing foils shows good improvement in CRs as lists continue, suggesting a gradual adjustment to the large number of confusing foils. In addition, current list targets show a decrease in Hs. In the modeling section we show how the increase in CRs occurs, and show how the decrease in Hs is both a natural result of a switch to the use of recall to reject.

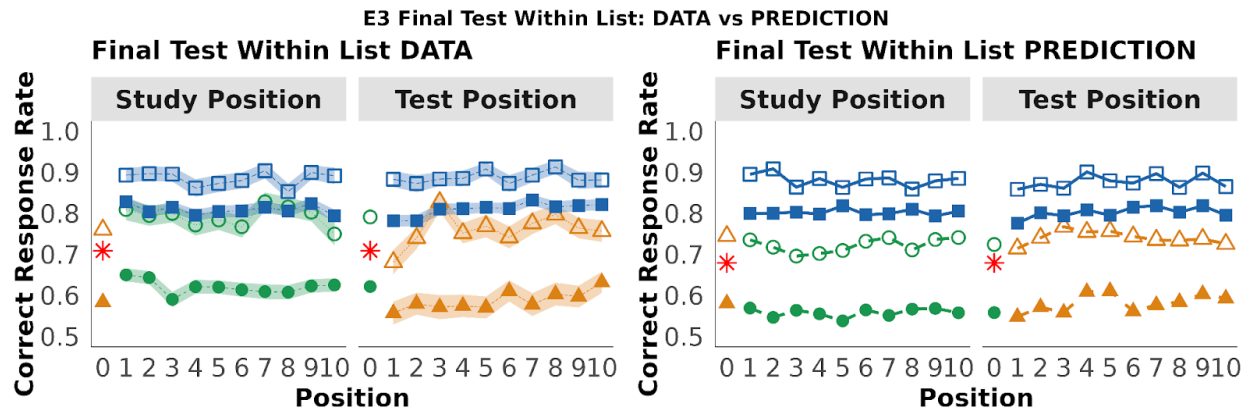


Figure 9.1.a.b and 9.2.a.b. 9.1: Correct response rates during final testing as a function of initial list study position (a) and test position (b). 9.2: Model predictions. **asterisks** = CRs for final test foils; **filled blue squares** = Hs for initial targets, not used as foils on next list; **unfilled blue squares** = Hs for initial list targets used as foils in next list; **filled orange triangles** = Hs for initial list foils not used as foils again on the next list; **unfilled orange triangles** = Hs for initial list foils used as foils again on next list; **filled green circles** = Hs for initial list pictures studied-only; **unfilled green circles** = Hs for initial list targets used as foils on next list; **Position 0** represents items without a study or test position. **Error ribbons** represent ± 1 standard error of the mean.

Within List Study and Test Position Effects During Final Testing

~~Initial list study position effects:~~

Figure 9.1.a-b. shows that final test pictures give higher hits in final testing the more times they are seen during the initial lists, when analyzed either by study or test position within the initial lists. There is little effect of list study or test position: (there being no obvious account for the low performance for list one for foils on one list used as foils on the next list, we assume that this finding is anomalous, or due to statistical variation, and do not try to explain it). As was the case for Exp. 1, the absence of study position effects in final testing suggests that the (rather small) serial position effects seen in initial testing were due to some sort of local context effects that are temporary. Given that the study position effects in initial testing did not persist to final testing, we decided not to include them in the modeling: They were not judged important enough to justify the additional complexity in the model.

Final testing performance as a function of initial list test position:

The within list test position effects shown in Figure 7 for targets and foils (targets increasing and foils decreasing across the test positions) do not appear in final testing. It is natural to assume therefore

that the Figure 7 results were not due to a factor that altered storage strengths. In the modeling section the effects in Figure 7 are therefore ascribed to ????????

Analysis of final test performance as a function of initial study position revealed minimal effects of study position. The baseline linear trend (for ST(n) items that also appeared as foils on list n+1) showed no significant effect, $b = -1.57$, $SE = 0.84$, $z = -1.86$, $p = .062$, 95% CI [-3.22, 0.08]. Item-type-specific linear trends showed significant negative slopes for SO(n) items that appeared only once (estimated marginal $b = -5.89$, interaction $b = -4.32$, $SE = 0.99$, $z = -4.37$, $p < .001$, 95% CI [-6.25, -2.38]) and SO(n) items that later appeared as foils on list n+1 (estimated marginal $b = -15.83$, interaction $b = -14.26$, $SE = 0.69$, $z = -20.61$, $p < .001$, 95% CI [-15.62, -12.91]), as well as for ST(n) items that appeared only once (estimated marginal $b = -4.30$, interaction $b = -2.73$, $SE = 0.70$, $z = -3.87$, $p < .001$, 95% CI [-4.11, -1.35]). Significant item type differences emerged in overall performance. ST(n) items that also appeared as foils on list n+1 showed the highest performance ($M = 89.5\%$, logit intercept $b = 2.16$, $SE = 0.07$, $z = 32.90$, $p < .001$, estimated marginal mean = 2.16, $SE = 0.07$), substantially outperforming all other categories. ST(n) items that appeared only once also showed high performance ($M = 83.0\%$, $b = 0.62$, $SE = 0.02$, $z = 25.79$, $p < .001$, EMM = 1.59, $SE = 0.05$). SO(n) items that later appeared as foils on list n+1 showed moderate performance ($b = 0.52$, $SE = 0.04$, $z = 12.43$, $p < .001$, EMM = 1.50, $SE = 0.06$), while SO(n) items that appeared only once showed substantially lower performance ($M = 56.7\%$, $b = -0.40$, $SE = 0.02$, $z = -19.34$, $p < .001$, EMM = 0.57, $SE = 0.05$). TO(n) items that appeared only once showed the lowest performance ($b = -0.60$, $SE = 0.03$, $z = -19.11$, $p < .001$).

—————~~Initial list test position effects:~~

—————~~Figure 9.2 shows that final test pictures give higher hits in final testing the more times they are seen in initial lists. There is little effect of initial test position. These results will be explained in the modeling section.~~

Analysis of final test performance as a function of initial test position revealed minimal effects of test position. The baseline linear trend (for ST(n) items that also appeared as foils on list n+1) was strongly positive, $b = 27.04$, $SE = 1.00$, $z = 26.94$, $p < .001$, 95% CI [25.07, 29.01], indicating that items tested later in the initial lists showed better final test performance. ST(n) items that appeared only once showed no additional linear trend beyond the baseline (estimated marginal $b = 26.75$, interaction $b = -0.29$, $SE = 1.25$, $z = -0.23$, $p = .817$, 95% CI [-2.73, 2.16]). TO(n) items that appeared twice (on list n and then as foils on list n+1) showed a negative deviation from the baseline trend (estimated marginal $b = 12.95$, interaction $b = -14.09$, $SE = 0.83$, $z = -17.07$, $p < .001$, 95% CI [-15.71, -12.48]), while TO(n) items that appeared only once showed a smaller negative deviation (estimated marginal $b = 18.86$, interaction $b = -8.18$, $SE = 0.99$, $z = -8.27$, $p < .001$, 95% CI [-10.11, -6.24]).

Item type differences revealed a clear performance hierarchy. ST(n) items that also appeared as foils on list n+1 showed the highest accuracy ($M = 90.4\%$, logit $b = 1.03$, $SE = 0.05$, $z = 19.79$, $p < .001$, EMM = 2.05, $SE = 0.07$), followed by ST(n) items that appeared only once ($M = 81.9\%$, $b = 0.44$, $SE = 0.03$, $z = 17.68$, $p < .001$, EMM = 1.47, $SE = 0.05$). TO(n) items that appeared twice showed moderate performance ($b = 0.18$, $SE = 0.04$, $z = 4.35$, $p < .001$, EMM = 1.20, $SE = 0.06$), while TO(n) items that appeared only once showed substantially lower performance ($M = 29.4\%$, $b = -0.73$, $SE = 0.03$, $z =$

-22.69, $p < .001$, EMM = 0.29, SE = 0.05). Pairwise comparisons confirmed all differences were significant (all $ps < .001$).

Final Test order effects:

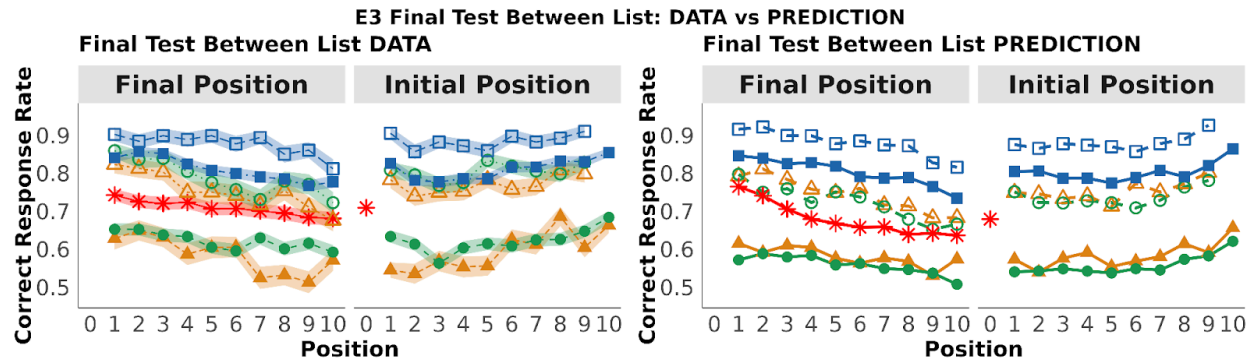


Figure 10.1.a.b and 10.2.a.b: 10.1.a: Final test accuracy as a function of final test position; the 492 final test trials are divided into 10 temporal bins (bins 1-8 contained 49 items each; bins 9-10 contained 50 items each) based on presentation order. 10.1.b: Final test accuracy as a function of initial list position. 10.2: Model predictions. (**blue squares** = targets, **orange triangles** = foils, **green circles** = studied-only, **solid shapes** = non-repeated items, **open shapes** = repeated items), with the addition of **red stars** representing **final foil items**. **Error ribbons** represent ± 1 standard error of the mean.

Figure 10.1.a shows strong evidence of output interference (OI). This finding is consistent with that from the random test condition of Exp. 1, and with the various demonstrations of strong OI by Criss, Malmberg, Shiffrin and others (discussed in the introduction) for recognition study followed by a sufficiently long series of tests. The accuracy of the different picture types generally follows the pattern seen in Exp. 1: Higher accuracy for pictures seen more times in the initial ten lists.

Across final test positions, all item types showed significant linear declines in accuracy, indicating strong output interference. The baseline decline (for final test foils) was substantial, $b = -30.78$, $SE = 1.27$, $z = -24.31$, $p < .001$, 95% CI [-33.27, -28.30]. This pattern varied significantly by item type. Current-list items showed varying decline rates. ST(n) items that also appeared as foils on list $n+1$ showed an additional decline (estimated marginal $b = -46.51$, interaction $b = -15.73$, $SE = 1.97$, $z = -8.00$, $p < .001$, 95% CI [-19.58, -11.87]), while ST(n) items that appeared only once showed a steeper additional decline (estimated marginal $b = -59.93$, interaction $b = -29.15$, $SE = 1.29$, $z = -22.66$, $p < .001$, 95% CI [-31.67, -26.63]). SO(n) items that appeared only once showed a positive deviation from baseline (estimated marginal $b = -25.00$, interaction $b = 5.78$, $SE = 1.04$, $z = 5.54$, $p < .001$, 95% CI [3.74, 7.83]), while SO(n) items that later appeared as foils on list $n+1$ showed the steepest decline (estimated marginal $b = -80.25$, interaction $b = -49.47$, $SE = 1.88$, $z = -26.27$, $p < .001$, 95% CI [-53.16, -45.77]). TO(n) items that appeared twice showed a substantial additional decline (estimated marginal $b = -73.13$, interaction $b = -42.34$, $SE = 1.51$, $z = -28.11$, $p < .001$, 95% CI [-45.30, -39.39]), while TO(n) items that appeared only once showed a moderate additional decline (estimated marginal $b = -56.94$, interaction $b = -26.15$, $SE = 1.78$, $z = -14.68$, $p < .001$, 95% CI [-29.65, -22.66]).

Each item type showed varying quadratic trends. The shared quadratic term was not significant, $b = 1.59$, $SE = 1.01$, $z = 1.58$, $p = .113$, 95% CI [-0.38, 3.56]. Item-type-specific quadratic trends varied substantially. ST(n) items that also appeared as foils on list n+1 showed a strong negative quadratic trend (estimated marginal $b = -30.57$, interaction $b = -32.16$, $SE = 1.36$, $z = -23.60$, $p < .001$, 95% CI [-34.83, -29.49]), indicating an inverted U-shape with particularly poor performance in the middle portion of the test. ST(n) items that appeared only once showed a positive quadratic trend (estimated marginal $b = 8.18$, interaction $b = 6.59$, $SE = 1.42$, $z = 4.63$, $p < .001$, 95% CI [3.80, 9.38]). SO(n) items showed contrasting quadratic patterns: those that later appeared as foils on list n+1 showed a positive quadratic trend (estimated marginal $b = 14.50$, interaction $b = 12.91$, $SE = 1.40$, $z = 9.23$, $p < .001$, 95% CI [10.17, 15.65]), while those that appeared only once also showed a positive quadratic trend (estimated marginal $b = 6.80$, interaction $b = 5.20$, $SE = 1.31$, $z = 3.97$, $p < .001$, 95% CI [2.64, 7.77]). TO(n) items that appeared only once showed a marginal positive quadratic trend (estimated marginal $b = 7.84$, interaction $b = 6.25$, $SE = 3.57$, $z = 1.75$, $p = .080$, 95% CI [-0.76, 13.26]).

Averaged across final test positions, planned pairwise comparisons with Tukey correction revealed a clear performance hierarchy (all pairwise differences $p < .001$): ST(n) items that also appeared as foils on list n+1 showed the highest performance ($M = 89.7\%$, $EMM = 2.17$, $SE = 0.07$), followed by ST(n) items that appeared only once ($M = 82.9\%$, $EMM = 1.58$), final test foils ($M = 72.7\%$, $EMM = 0.98$), TO(n) items that appeared twice ($EMM = 1.26$), SO(n) items that later appeared as foils on list n+1 ($EMM = 1.46$), SO(n) items that appeared only once ($M = 54.9\%$, $EMM = 0.55$), and TO(n) items that appeared only once ($M = 38.1\%$, $EMM = 0.38$).

Initial List order effect

Figure 10.1.b shows final test accuracy as a function of the initial list position. The results are similar to the Exp. 1 results shown in Figure 5.1.b for the random test group. There is some slight recency (higher accuracy for pictures seen on the more recent lists). 10.2: Model predictions.

The baseline linear trend (for ST(n) items that also appeared as foils on list n+1) was not significant, $b = 1.83$, $SE = 1.11$, $z = 1.65$, $p = .098$, 95% CI [-0.34, 4.01]. However, item-type-specific linear trends showed substantial variation. TO(n) items that appeared only once showed the strongest linear improvement (estimated marginal $b = 58.86$, interaction $b = 57.03$, $SE = 0.66$, $z = 86.33$, $p < .001$, 95% CI [55.73, 58.32]), indicating dramatically better retention for items from more recent lists. SO(n) items that later appeared as foils on list n+1 also showed strong improvement (estimated marginal $b = 22.50$, interaction $b = 20.67$, $SE = 0.75$, $z = 27.46$, $p < .001$, 95% CI [19.19, 22.14]), as did TO(n) items that appeared twice (estimated marginal $b = 17.37$, interaction $b = 15.53$, $SE = 0.73$, $z = 21.37$, $p < .001$, 95% CI [14.11, 16.96]). ST(n) items that appeared only once showed moderate improvement (estimated marginal $b = 9.76$, interaction $b = 7.93$, $SE = 0.77$, $z = 10.35$, $p < .001$, 95% CI [6.43, 9.43]), while SO(n) items that appeared only once showed a small negative trend (estimated marginal $b = 0.27$, interaction $b = -1.56$, $SE = 0.80$, $z = -1.96$, $p = .050$, 95% CI [-3.12, -0.001]).

Each item type also showed significant quadratic acceleration, indicating accelerating improvement over positions. The shared quadratic term was significant, $b = 1.57$, $SE = 0.58$, $z = 2.72$, $p = .007$, 95% CI [0.44, 2.71]. Item-type-specific quadratic trends varied substantially. ST(n) items that appeared only once showed the strongest positive quadratic acceleration (estimated marginal $b = 33.79$,

interaction $b = 32.21$, $SE = 0.63$, $z = 50.89$, $p < .001$, 95% CI [30.97, 33.45]), followed by SO(n) items that appeared only once (estimated marginal $b = 29.93$, interaction $b = 28.35$, $SE = 1.07$, $z = 26.53$, $p < .001$, 95% CI [26.26, 30.45]). TO(n) items that appeared only once showed a smaller but significant positive quadratic trend (estimated marginal $b = 7.16$, interaction $b = 5.58$, $SE = 0.76$, $z = 7.39$, $p < .001$, 95% CI [4.10, 7.06]). In contrast, SO(n) items that later appeared as foils on list $n+1$ showed a strong negative quadratic trend (estimated marginal $b = -44.15$, interaction $b = -45.72$, $SE = 1.08$, $z = -42.36$, $p < .001$, 95% CI [-47.84, -43.61]), as did TO(n) items that appeared twice (estimated marginal $b = -29.92$, interaction $b = -31.50$, $SE = 1.18$, $z = -26.58$, $p < .001$, 95% CI [-33.82, -29.17]).

Averaged across all list positions, planned pairwise comparisons revealed a clear performance hierarchy (all $ps < .001$): ST(n) items that also appeared as foils on list $n+1$ showed the highest performance ($M = 89.6\%$, $EMM = 2.16$, $SE = 0.07$), followed by ST(n) items that appeared only once ($M = 83.3\%$, $EMM = 1.61$), TO(n) items that appeared twice ($EMM = 1.32$), and SO(n) items that later appeared as foils on list $n+1$ ($EMM = 1.16$). SO(n) items that appeared only once ($M = 59.5\%$, $EMM = 0.59$) and TO(n) items that appeared only once ($M = 28.8\%$, $EMM = 0.29$) showed the lowest performance.

Participant Performance

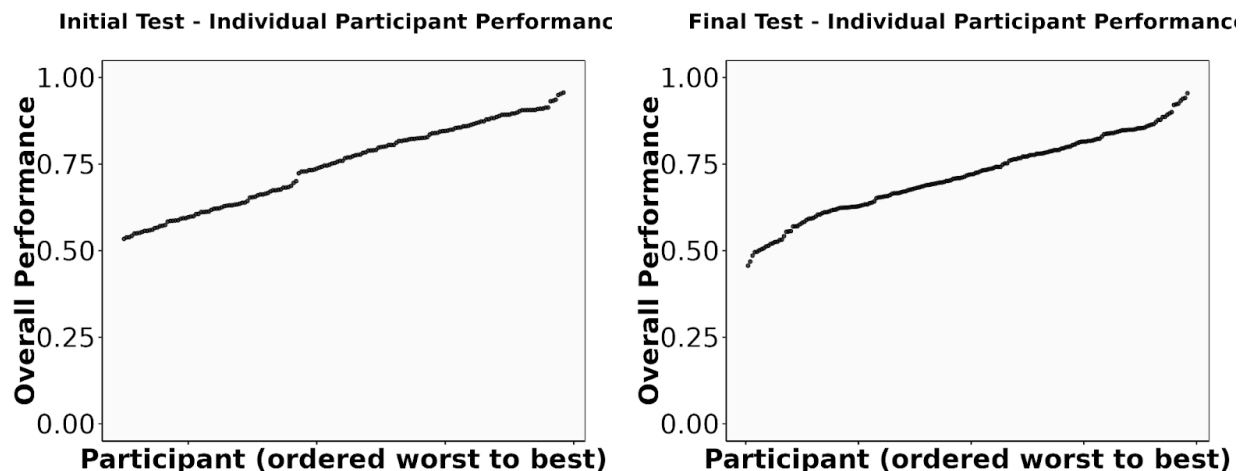


Figure 11.1.2: Experiment 2 individual differences: 11.1: Participant performance on the initial ten lists ordered by accuracy; 11.2: Participant performance during final testing ordered by accuracy.

Figure 11 shows a very large range of individual differences in Exp. 2, similar to those shown for Exp. 1. In contrast to Exp. 1, the data for the first ten lists exhibits quite low accuracy, and there is not a sharp decrease to the final testing level. These results are due to the very high false alarm rates for the confusing foils in Exp 2 during the testing of the first ten lists.

We repeat the reasons why these results are important for modeling: All the data predicted are an average of individual data, and hence a mixture of widely varying results. Under these circumstances, and because the data per condition is too sparse to allow separate modeling of individuals, we cannot expect precise quantitative predictions. Thus the goal of modeling is qualitative: We will produce a model that

can predict the trends in the data, and will try to choose parameter values so that the predictions are not too discrepant quantitatively.

Modeling Experiments 1 and 2

Overview:

The starting point is the simple three parameter REM model (bREM) of Shiffrin and Steyvers (1997) that has proved remarkably capable of capturing the major trends of recognition studies (e.g. 500+ conditions in the research of Mohamed et al., in press). bREM is missing many processes known or very likely to play important roles in recognizing. Thus the present studies were aimed to provide data helping to augment bREM by incorporating important missing processes. bREM was described in the introduction and need not be repeated here.

It is important to repeat that the goal of the modeling is qualitative, not a model that predicts the data as precisely as possible. There are enormous differences in performance among participants seen in Figures 6 and 11. The model predicts data that is a complex mixture averaged across not just across participants but across pictures, and across other variables that might affect performance. At best such a model is a crude approximation, so we aim to predict the major observed trends rather than quantitative details.

The basic assumptions of bREM are used in the present modeling, with some elaborations described in the following sections. To be specific, each picture is represented by 23 content features (termed T), and two types of context features (delineated shortly): 28 CC features and 28 UC features. Each feature value for each picture is sampled independently from a geometric distribution with parameter value $g = 0.3$. Each feature of a picture studied is stored (or not) with probability $u^* = 0.4$. If stored the probability the value will be stored correctly is $c = 0.85$ (Exp. 1) and 0.70 (Exp. 2)

Augmenting REM with processes missing from bREM.

General background concerning bREM and some processes that bREM is missing has been covered in the introduction. Here we give more detail, and the way that the model implements these processes.

Context features:

1. Context features used to activate traces:

As suggested by Shiffrin and Steyvers (1997) CC and UC context features are used alone, without content features, in a first stage of retrieval. This stage determines which traces are sufficiently similar in context to the context in the retrieval probe to become activated and then passed to the second stage. This two stage approach is meant as an easy-to-implement approximation to a single stage system in which context and content are used together to probe memory (as was used in the dynamic model of Cox & Shiffrin, 2017). The first stage uses the regular REM machinery to determine trace likelihood ratios. There is a threshold with value $t_t = 100$. Traces with first stage likelihood ratios greater than 100 are passed to the second stage. The second stage uses only content (T) features and compares the T features of the test picture to the T features in each activated trace, as in bREM.

2. Context feature types:

There could be a wide variety of context features for each stage and each condition in the present studies, but doing so would have added too many parameters and processes to explain what the data show, and to justify the increased complexity. Thus we distinguish explicitly two types of context features: context that changes during the first ten lists, termed ‘changing context’, or CC-(28 in number), and context that does not change during the first ten lists, termed ‘unchanging context, or UC. All pictures are represented with 25 content features -(T = 25), 28 changing context features (CC = 28), and 28 unchanging context features (UC = 28). In different conditions and studies not all these features are necessarily used for study???? and test. (also 28 in number).

3. Context storage, retrieval, and change::

Storage:

A}During study in every condition of Experiment 1 every context and content feature is stored with the bREM parameters $g = 0.3$, $c = 0.7$, $u = 0.4$. During study in every condition of Experiment 2 every context and content feature is stored with the bREM parameters $g = 0.3$, $c = 0.8$, $u = 0.4$.

Distortion and Recovery:

We assume that the context and content features available at the start of testing are distorted by the change from study to test, but that both types of features

Context Change:

1] *During Study of each list*: Changing context (CC) does not change during study. CC changes could have been assumed during study of each initial list, thereby accounting for the serial position effects found in the data. However, we decided the added model complexity would not justify adding such assumptions, especially because the CC changes would simply be made to match the serial position effects observed.

~~B]~~ Context changes:

~~21]. CC changes~~ *Between study and test for each list*. Each CC feature changed ~~Between Study and Test~~ from its current value to a sample from the geometric distribution. The change occurs with probability $\delta_{BST} = 0.2396$ (Exp. 1) and 0.307 (Exp. 2). The parameter value of the geometric distribution was $g = 0.3$.

~~23]. CC~~ *Recovery during testing*: The change from study to test is assumed to be large enough that recovery of some features occurs over the first several tests. At each test each feature value has a probability of reverting to its value at study. That of ~~reco~~ Recovery probability of $\delta_R = 0.3$ (common to both experiments) of reverting to the value at the end of study.

3. **CC changes between test of a list and study of the next list**: Each CC feature changed from its current value to a sample from the geometric distribution (again with $g = 0.3$). The change probability **Between List** was $\delta_{BL} = 0.456$ (common to both experiments).

4. **CC changes between initial lists and the final test**. Following completion of all ten lists, each CC feature changed from its current value to a sample from the geometric ($g = 0.3$). The probability of change **Between Initial and Final test** was $\delta_{BIF} = 0.08$ (Exp. 1) and 0.18 (Exp. 2) .

5. **CC changes during final testing**. CC features changed every 42 tests from the current value to a choice from the geometric (again with $g = 0.3$). The

probability of change **Between Final testing** was $\delta_{BF} = 0.02$ (common to both experiments).

D] Unchanging context:.

1. **UC changes when final testing begins:** Each UC feature changed from its current value to a sample from the geometric ($g = 0.3$). The probability of change was $\delta_{BIF} = 0.08$ (Exp. 1) and 0.18 (Exp. 2).

2. **UC changes during final testing:** CC features changed every 42 tests from the current value to a choice from the geometric (again with $g = 0.3$). The probability of change was $\delta_{BF} = 0.02$.

E] Retrieval with CC and UC features: During the first ten lists of both Experiments 1 and 2, the first stage of retrieval uses the current values of all the CC and UC features to probe memory and produce traces that are activated: The traces activated are those with stage 1 likelihood ratios > 100 . The experiments require a focus on the most recent list studied, so we assume that the retrieval probe in stage 1 uses all 28 CC features, but only the first 13 of the 28 UC features. The situation changes for final testing, because all traces from earlier lists are targets. Thus stage 1 retrieval in final testing uses 28 UC features and the first 8 of the 28 CC features.

Storage of traces during testing:

Testing an item is an event and like all events gets stored. Every test event should therefore produce storage, storage of both content and context features. We assume that each study or test event for a picture produces a trace stored as in bREM, with parameters $u = 0.4$; $c = 0.85$ (Exp. 1), 0.70 (Exp. 2); $g = 0.3$. Thus::

- a] study of a picture without any subsequent tests produces a single trace;
- b] that a picture tested just once (as a new foil) produces a single test trace;
- c] that a picture studied and tested on the same list, but not tested again, produces two traces, one for study, one for test;
- d] that a picture studied, not tested on that same list, but then presented as a foil on the next list has two traces stored, one for study, one for test.
- e] that a picture tested as a foil and then tested again as a foil on the next list, has two traces stored, one for each test;

f) that a picture studied and then tested on that same list and then tested as a foil on the next list has three traces stored, one for study and one for each test.

These storage assumptions superficially deviate from previous modeling of list-strength effects by Shiffrin and Steyvers (1997, 1998) who argued that new traces were not stored most of the time when an item was studied several times on the same list. However, we are now discussing storage of test traces, a rather different process. What is the amount of storage during testing is an open question answered in part by a tested picture's performance in final testing. However, note that the final testing performance level for pictures tested in the first ten lists is due to several other factors than storage of a new trace, the most important being strengthening of an existing trace, described next. Storage of test traces adds traces that pass stage 1 filtering (due to context similarity), are activated, and then take part in stage two (based on their content features). How many are activated (beyond the stage 1 threshold of $t_i = 100$ depends on context change and the numbers of UC and CC features used to probe memory. Whatever the numbers, the additional activated traces act like an increase in list-length and decrease performance as testing continues, thus a factor that produces output interference (OI). This factor tends not to be very large, however, and a different factor may be as important, or more important, in producing OI: the result of strengthening of existing traces, discussed next.

Strengthening existing traces:

As discussed by Shiffrin and Steyvers (1997) strengthening existing traces during study was essential to allow REM to predict list-strength effects. The idea explained briefly: When an item is studied additional times (massed or spaced), its trace tends to be sampled in a recall process operating in parallel with computation of odds. When sampled that trace is strengthened by making its features better match the current test presentation. That produces differentiation – makes the trace less similar to other items to be tested later, thereby improving recognition of those to be tested items.

Similar processes operate during testing. When items are repeatedly tested, REM familiarity (odds) is calculated, and in parallel a trace is sampled from those currently activated. The probability of sampling a trace is the ratio of its likelihood ratio to the sum of likelihood ratios of all activated traces. Sampling tends to produce choice of the strongest trace, so when a target is tested, its trace is often the one sampled. If the sampled trace has a likelihood value greater than a recall threshold of $t_r = 0.2$ (Exp. 1) and 0.3 (Exp. 2), we assume that the participant makes the default assumption that the trace sampled is indeed that of the test item, and therefore strengthens it by filling in trace features found in the test item that are missing

from the trace, and by correcting trace features that do not match those in the test item. To be specific, a missing feature in the trace and a trace feature not matching the test item is replaced by the feature in the test item with the usual storage rules, based on *u*, *c*, and *g*.

Strengthening a trace has the direct effect of improving later retrieval when a later test of a target causes sampling of that trace. However, strengthening has two *opposing* effects on tests of other items tested later:

1] The first effect is differentiation: In most cases, strengthening a trace makes it less similar to other stored traces (the exception not relevant for the present studies occurs when the item tested later is highly similar to the one whose trace was strengthened, as happens for example if a word is strengthened and later that word's plural is tested). A later test of some other item will therefore match the strengthened trace less well, reducing that trace's likelihood ratio, and reducing familiarity [odds]. The net result will be an increase in accuracy for a future test of a different item. This process by itself produces a result opposite of OI.

2] There is a different effect of strengthening that in many cases will produce the opposite effect, an increase in OI. In many studies there are a good number of false alarms when a foil is tested. When a false alarm occurs to a new foil not seen earlier in the experiment, the trace sampled cannot be the one produced by the test item. Instead the sampled trace will be that of some item seen earlier, often a target that has yet to be tested. When a target trace is sampled, adding to it the features of the foil currently being tested will degrade its quality. When the item that did produce that trace is later tested, performance will drop (its hit probability will drop), and hence OI will increase. In a typical recognition study with foils tested half the time, and with a good number of false alarms, the balance of the two opposing effects can produce a net effect of output interference [OI]; our simulations have shown that the OI produced by strengthening can be a larger cause of OI than that produced by storage of test traces.

The use of recall in recognition:

The use of recall in recognition studies has been studied and modeled for many years, as discussed in the introduction. The process of strengthening, either for repetitions during study, or during testing, is based on recall operating in parallel with calculation of familiarity (i.e. in parallel with computation of REM odds). However bREM and its successors have until now used familiarity (odds) for decision making, so recall has affected decisions indirectly. The success of bREM and its successors using only odds for decisions is largely due to the foils that are used in typical recognition studies: These are typically new items not seen earlier in the experiment. In

such designs (like the present Exp.1), any item reasonably familiar is a target. Further, an item with high familiarity will produce a trace likely to be recalled, so a decision based on familiarity will tend to be the same as one produced by examination of a recalled trace. This correlation of predictions made by the two decision processes explains why use of familiarity alone suffices to predict recognition.

There is nonetheless a long history of studies (*see the introduction*) making a strong case that recall plays an important role in recognizing, even in standard experimental designs. The results of the present Experiment 2 makes this case even more strongly: In Experiment 2, when familiarity is well above the level needed to respond old, it is often the case that the high familiarity derives from the test item having appeared on the prior list, not the current list. In such cases the test item is a foil that requires a new decision. Thus a decision based on familiarity alone would produce a high number of false alarms (which it does), but more critically would predict the wrong ordering of false alarms across the conditions. Thus to model the observed results, we assume that recall of a trace occurs in parallel with computation of familiarity. When that trace is found to contain information that the trace was produced by appearance on a list other than the current list, then a decision can be made to respond new despite the high familiarity of the test picture. This is the process known as ‘recall to reject’.

The existence of origin information in a trace should vary with the history of the picture used to produce that trace. E.g. A picture studied and tested should produce a trace with better origin information than the trace of a picture studied and not tested. Given that a trace is recalled and origin information is found, should that information be used to alter responding from that based on familiarity? We assume the origin information is noisy and sometimes wrong. Given the existence of error we assume there is a probability of using origin information found in a trace, a probability that grows as lists continue and as the participants adapt their coding to the presence of so many confusing foils from the prior list.

To implement these processes in the modeling, we assume existence of a special feature called **Z**: $Z = 0$ indicates that the trace contains information that the picture producing the trace had *not been seen on a previous list*. $Z = 1$ indicates that the trace contains information that the picture producing the trace had *been seen on a previous list*. The Z information is noisy and can be wrong. $p[Z=0] + p[Z=1] = 1$.

The Z values found in traces that are sampled and recalled when a picture is tested:

These are stipulated for lists 2 through 10. What they are on list 1 is irrelevant because familiarity is used to decide on list 1. We assume that the origin information in a sampled trace is used to reject only for traces with high enough likelihood ratios to be recalled. For simplicity we

assume further that this process only operates when the trace sampled was produced by study or test of the currently tested picture.

A] A trace produced by a picture studied on the current list but not seen on a prior list. Let $ku(j)$ be the probability that $Z = 1$ for list j , $j > 1$. In Exp. 2 a target was never on a previous list, so $Z = 1$ indicates an error in coding origin information.

B] A trace produced by a picture studied but not strengthened on the prior list, then tested as a foil on the current list. Let $ks(j)$ be the probability that $Z = 0$ for list j , $j > 1$. Thus $Z = 0$ indicates an incorrect coding that this trace was produced by a picture on the prior list

C] A trace produced by a picture not studied but tested on the prior list and then tested as a foil on the current list. Let $kt(j)$ be the probability that $Z = 0$ for list j , $j > 1$. Thus $Z = 0$ indicates an incorrect coding that this trace was produced by a picture on the prior list.

D] A trace produced by a picture studied on a prior list and then strengthened by a subsequent test on some prior list, and then tested as a foil on the current list. Let $kb(j)$ be the probability that $Z = 0$ for list j , $j > 1$. Thus $Z = 0$ indicates an incorrect coding that this trace was produced by a picture on the prior list.

Note that the $k(j)$ terms all give probabilities of error in coding origin information. We assume that participants gradually adapt to the importance of coding origin information correctly, so the error probabilities drop list by list according to a function $e(j)$. Thus each $k-(j) = k-(2)e(j)$.

When is a decision made to recall-to-reject:

Consider a test that produces odds greater than the decision criterion $t_o = 0.176$. A trace is sampled that is a trace produced by study or test of the current test picture. Suppose its likelihood ratio is greater than the recall threshold $t_r = 0.3$. Let the probability that the Z value will be used to decide be $f(j)$ for list j , $j > 1$. $f(j)$ will increase list by list. This increase in use of origin information is prompted both by the decrease in the Z error probabilities, and the need to increase CRs for the many confusing foils.

The probability on list j of a hit to a target, or a correct rejection to the different kinds of foils, is difficult to predict without simulation. However, it can be seen that the drop in hits for target tests as lists continue is based on a switch from use of familiarity (which produces very high hit rates, as seen in the data on list 1) to recall (which is error prone and less efficient than familiarity). Why then does the participant switch? Without switching the false alarms to the many confusing foils would be enormously high, even higher than those observed. False alarms are high because the change in context between lists, and the use of some UC context features to probe memory causes activation of traces from prior lists. The high false alarm rates observed

for the confusing foils (i.e. low CR rates) make it necessary to assume processes that produce activation of prior list traces.

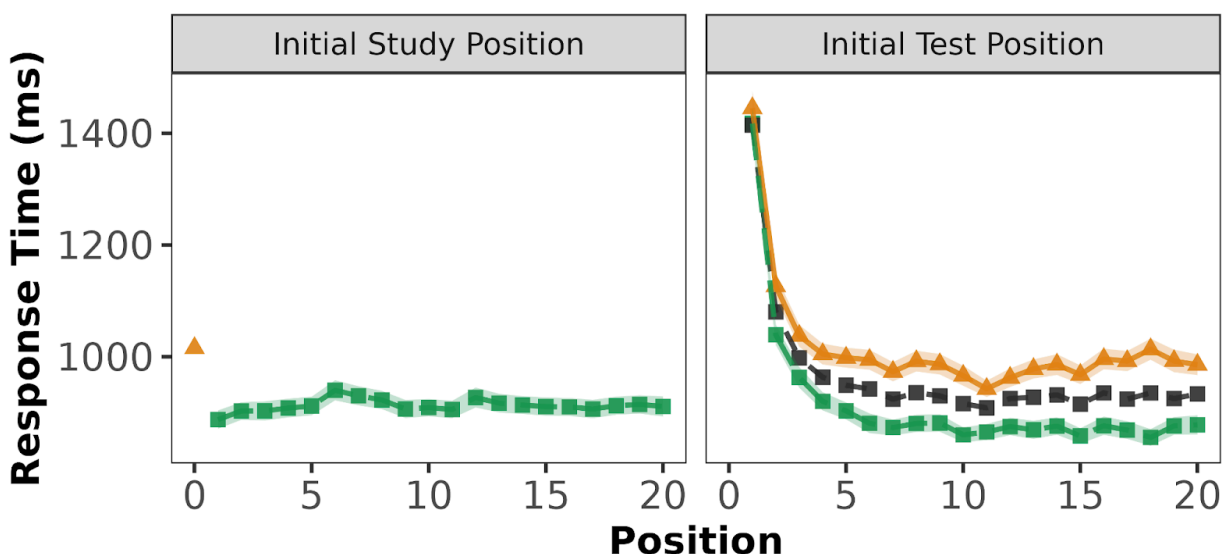
To sum up, the probability of responding old on list j will be determined by familiarity unless the trace sampled on that test is recalled. If the familiarity is above the decision threshold of 0.176, and if the sampled trace is recalled with a Z value of 1 and if a decision is made to use that origin information, then the participant will switch from an old response based on familiarity to a new response based on recalled information.

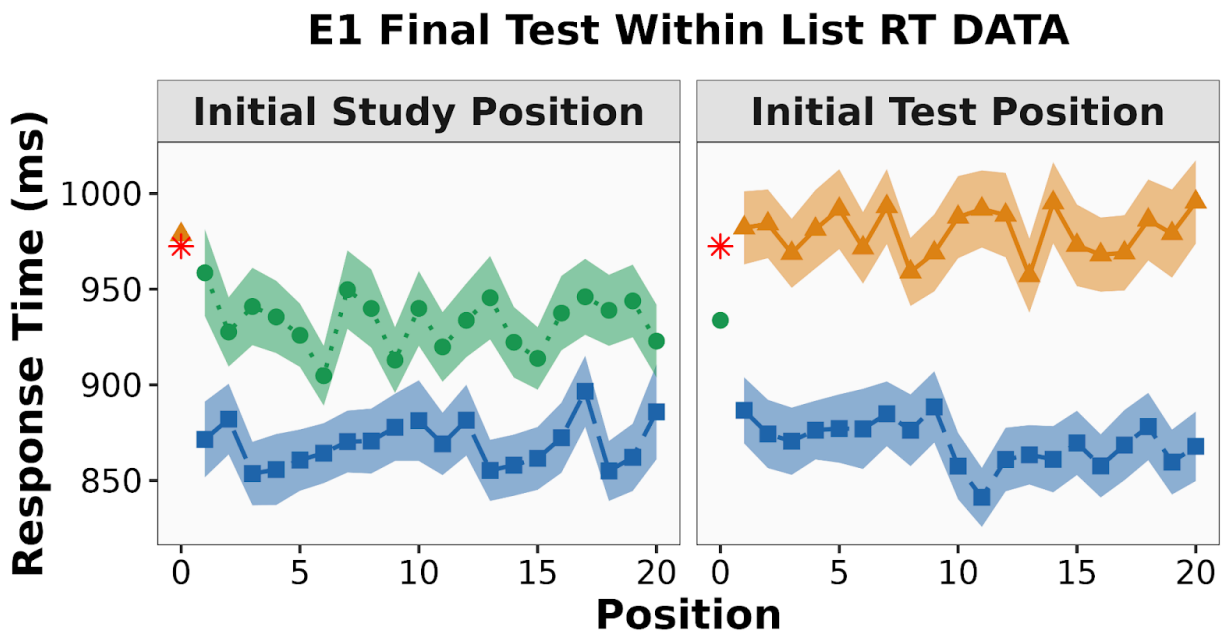
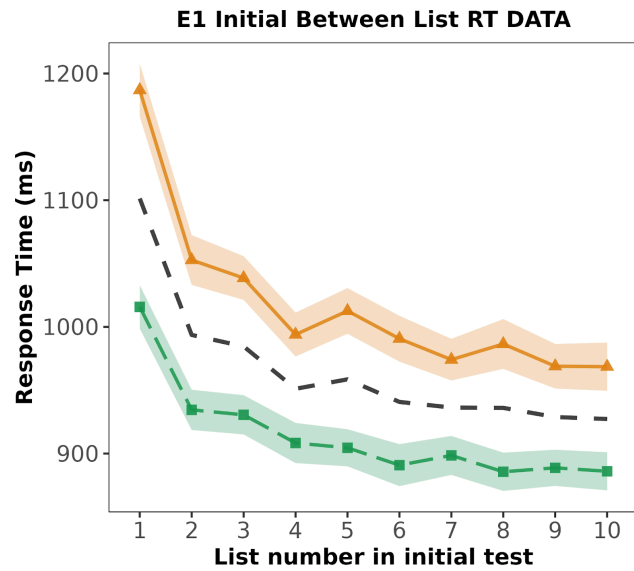
Processes important for recognition not modeled:

As discussed in the introduction we do not implement several processes important for recognition, including 1. **Associations between studied items:** Important but not added to the present modeling. 2. **Differences among content features:** All content features are modeled as if sampled from the same distribution of features. 3. **Dynamics of feature activation during storage and retrieval:** Different feature types surely are activated at different times (as modeled by Cox & Shiffrin, 2017) but such dynamics are not part of the augmented model.

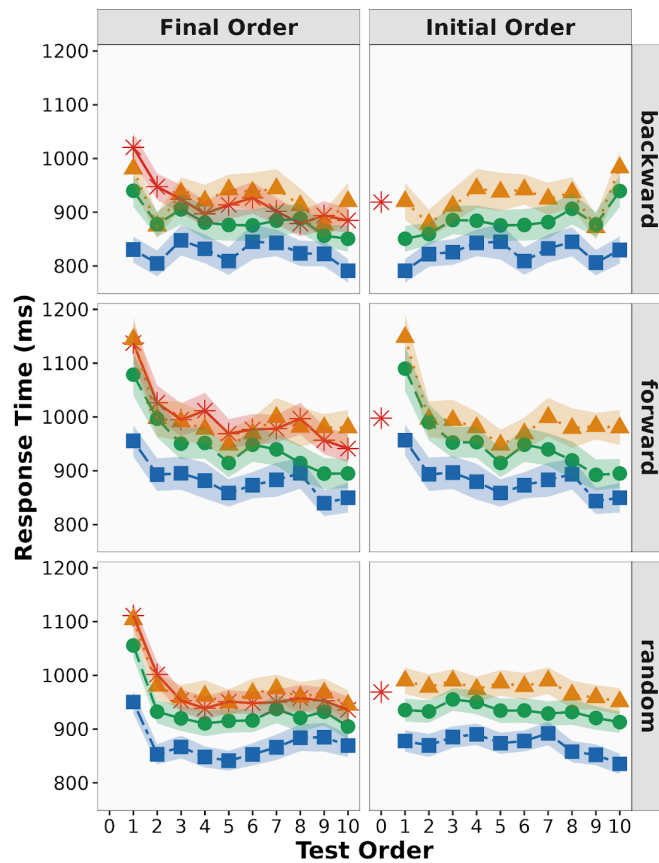
Appendix

E1 Initial Within List RT DATA



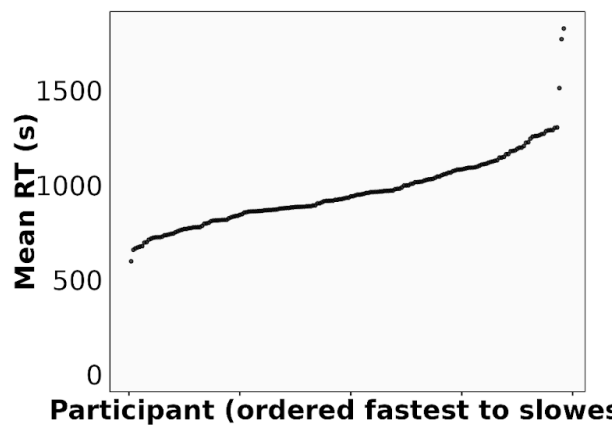


E1 Final Test Between List RT DATA

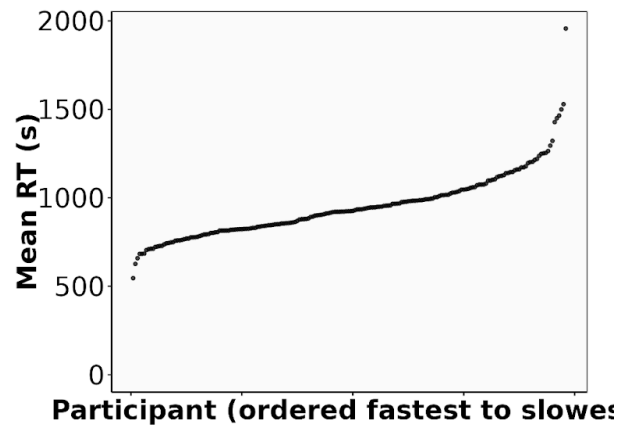


E1 Individual Participant Mean RT

E1 Initial Test - Individual Participant Mean RT



E1 Final Test - Individual Participant Mean RT



Retention of Slow Responders

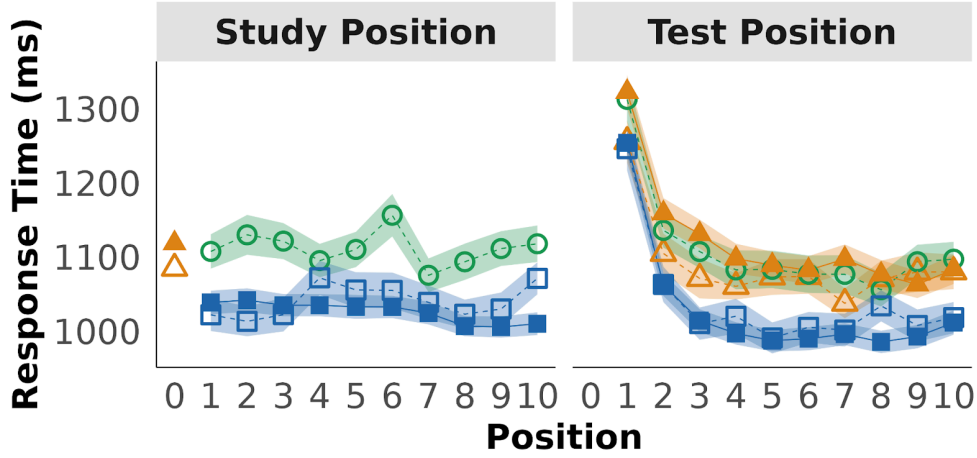
Visual inspection of individual reaction time (RT) plots revealed several participants with notably slower responses (median RTs approximately 2–3 standard deviations above the sample mean). These participants were retained in the final analyses because they satisfied all pre-specified inclusion criteria and demonstrated consistent, accurate performance.

As specified in the main text, exclusions were based on median RTs rather than means, using a conservative ± 3 SD cutoff. This approach minimizes the undue influence of a small number of unusually slow trials and better reflects participants' typical response latencies. Under this criterion, participants with median RTs ≤ 3 SD from the sample mean were retained. All such individuals exhibited accuracy well above chance (range = 0.64–0.93), stable response patterns across trials, and no evidence of fatigue or disengagement. Retaining these participants preserves genuine individual differences in processing speed (e.g., more deliberate or cautious response styles) and avoids biasing the sample toward unusually fast responders.

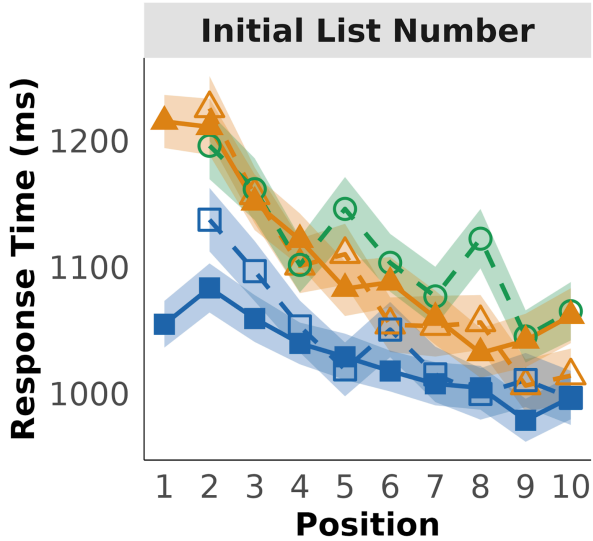
One participant exhibited a median RT of approximately 1,858 ms ($z = 6.21$) during the final test, exceeding the nominal exclusion threshold. This participant nonetheless maintained high task accuracy (0.73) and displayed stable performance across blocks, suggesting a consistently slow but attentive response style rather than random or inattentive behavior. Sensitivity analyses conducted with and without this participant yielded no qualitative changes to the results or statistical conclusions.

In sum, all retained participants either met the pre-specified inclusion criteria (median RT ≤ 3 SD) or demonstrated performance indicative of genuine task engagement. Excluding these individuals would have introduced an unnecessary sampling bias, while their inclusion did not alter the overall pattern of findings.

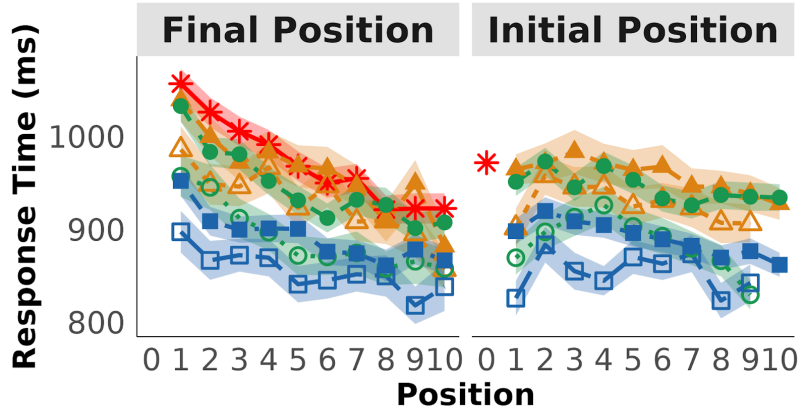
Exp. 2 Initial Test Within List RT DATA

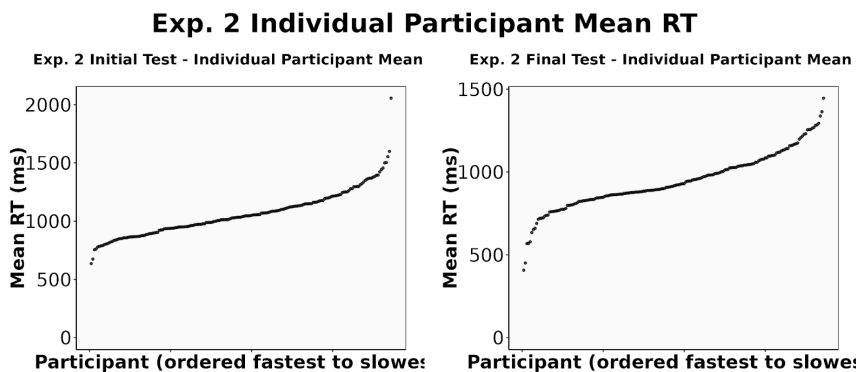
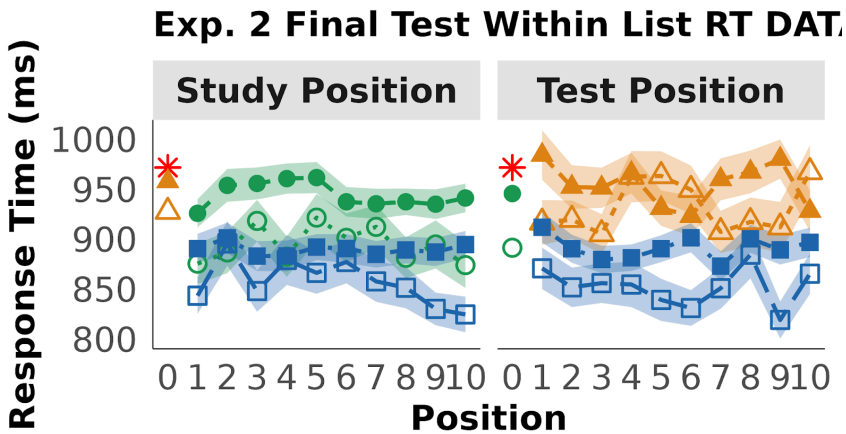


Exp. 2 Initial Between List RT



Exp. 2 Final Test Between List RT DATA





Retention of Slow Responders (Experiment 3)

As in Experiment 1, visual inspection of reaction time (RT) distributions in Experiment 3 revealed several participants with slower response profiles (median RTs $\approx$ 1,200–1,400 ms). Consistent with the pre-specified exclusion criteria, participants were retained if their *median* RTs were within 3 SD of the sample mean or marginally above this threshold when supported by strong accuracy and consistent engagement.

The slowest retained participant exhibited a median RT of 1,377 ms ($z = 3.09$) with 79.7% accuracy—only 0.09 SD above the 3 SD cutoff. This individual, together with other slow but accurate participants (accuracy range = 0.61–0.89), displayed stable RT distributions and no evidence of inattention or fatigue. Excluding these participants would have introduced bias against slower but deliberate responders. One participant ($z = 6.33$, accuracy = 0.56) was excluded in accordance with the criteria, demonstrating that the filtering procedure effectively removed genuine outliers.

Retention of these borderline cases reflects the same conservative and participant-inclusive rationale adopted in Experiment 1: slow responders were preserved when their data indicated genuine task

engagement, high accuracy, and consistent performance. Sensitivity analyses excluding these individuals did not alter the statistical or qualitative pattern of results.

Appendix B

Table 1

E1 Final Test Between List: Hits and Correct Rejections by Condition, Position Type, and Test Order

BACKWARD - Final Order

Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10
Foil	--	0.86 (0.02)	0.85 (0.02)	0.86 (0.02)	0.86 (0.02)	0.83 (0.02)	0.81 (0.02)	0.82 (0.02)	0.80 (0.02)	0.80 (0.02)	0.82 (0.02)
T_foil	--	0.90 (0.02)	0.76 (0.03)	0.65 (0.03)	0.64 (0.03)	0.63 (0.03)	0.61 (0.03)	0.60 (0.03)	0.55 (0.03)	0.58 (0.03)	0.57 (0.03)
T_nontarget	--	0.80 (0.03)	0.70 (0.04)	0.63 (0.04)	0.61 (0.04)	0.69 (0.03)	0.63 (0.03)	0.63 (0.03)	0.62 (0.03)	0.64 (0.04)	0.67 (0.03)
T_target	--	0.97 (0.01)	0.94 (0.02)	0.89 (0.02)	0.88 (0.02)	0.91 (0.02)	0.82 (0.03)	0.88 (0.02)	0.85 (0.02)	0.84 (0.03)	0.91 (0.02)

E1 Final Test Between List: Hits and Correct Rejections by Condition, Position Type, and Test Order

BACKWARD - Final Order

BACKWARD - Initial Order

Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10
Foil	0.83 (0.02)	--	--	--	--	--	--	--	--	--	--
T_foil	--	0.57 (0.03)	0.58 (0.03)	0.55 (0.03)	0.60 (0.03)	0.61 (0.03)	0.63 (0.03)	0.64 (0.03)	0.65 (0.03)	0.77 (0.03)	0.89 (0.02)
T_nontarget	--	0.67 (0.03)	0.64 (0.04)	0.62 (0.03)	0.63 (0.03)	0.63 (0.03)	0.69 (0.03)	0.61 (0.03)	0.63 (0.04)	0.70 (0.04)	0.81 (0.03)
T_target	--	0.91 (0.02)	0.84 (0.03)	0.85 (0.02)	0.88 (0.02)	0.82 (0.03)	0.91 (0.02)	0.88 (0.02)	0.90 (0.02)	0.94 (0.02)	0.97 (0.01)

FORWARD - Final Order

Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10
Foil	--	0.85 (0.02)	0.82 (0.02)	0.84 (0.02)	0.84 (0.02)	0.81 (0.02)	0.79 (0.02)	0.78 (0.02)	0.77 (0.02)	0.78 (0.03)	0.79 (0.02)
T_foil	--	0.72 (0.03)	0.66 (0.04)	0.66 (0.04)	0.63 (0.04)	0.65 (0.04)	0.66 (0.03)	0.67 (0.03)	0.70 (0.03)	0.65 (0.03)	0.67 (0.03)
T_nontarget	--	0.77 (0.03)	0.68 (0.03)	0.63 (0.03)	0.67 (0.04)	0.64 (0.04)	0.66 (0.04)	0.65 (0.04)	0.65 (0.04)	0.66 (0.04)	0.68 (0.04)
T_target	--	0.95 (0.01)	0.92 (0.02)	0.89 (0.02)	0.86 (0.02)	0.84 (0.02)	0.85 (0.02)	0.83 (0.03)	0.85 (0.03)	0.87 (0.03)	0.86 (0.03)

FORWARD - Initial Order

Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10
Foil	0.81 (0.02)	--	--	--	--	--	--	--	--	--	--
T_foil	--	0.71 (0.03)	0.66 (0.04)	0.65 (0.04)	0.63 (0.04)	0.65 (0.04)	0.66 (0.03)	0.67 (0.03)	0.70 (0.03)	0.65 (0.03)	0.67 (0.03)
T_nontarget	--	0.78 (0.03)	0.68 (0.03)	0.62 (0.03)	0.67 (0.04)	0.64 (0.04)	0.66 (0.04)	0.65 (0.04)	0.65 (0.04)	0.66 (0.04)	0.68 (0.04)
T_target	--	0.94 (0.01)	0.93 (0.02)	0.88 (0.02)	0.87 (0.02)	0.84 (0.02)	0.85 (0.02)	0.83 (0.03)	0.85 (0.03)	0.87 (0.03)	0.86 (0.03)

RANDOM - Final Order

Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10
Foil	--	0.87 (0.01)	0.84 (0.01)	0.83 (0.01)	0.84 (0.01)	0.83 (0.01)	0.83 (0.01)	0.84 (0.01)	0.83 (0.01)	0.82 (0.01)	0.81 (0.01)
T_foil	--	0.73 (0.02)	0.70 (0.02)	0.65 (0.02)	0.66 (0.03)	0.61 (0.02)	0.59 (0.02)	0.57 (0.03)	0.57 (0.02)	0.59 (0.03)	0.58 (0.03)
T_nontarget	--	0.72 (0.02)	0.70 (0.02)	0.71 (0.02)	0.68 (0.02)	0.63 (0.02)	0.63 (0.02)	0.63 (0.03)	0.62 (0.02)	0.63 (0.02)	0.64 (0.02)
T_target	--	0.94 (0.01)	0.93 (0.01)	0.88 (0.02)	0.87 (0.02)	0.88 (0.02)	0.88 (0.02)	0.86 (0.02)	0.85 (0.02)	0.86 (0.02)	0.84 (0.02)

RANDOM - Initial Order

Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10
Foil	0.84 (0.01)	--	--	--	--	--	--	--	--	--	--
T_foil	--	0.61 (0.02)	0.64 (0.02)	0.61 (0.02)	0.59 (0.02)	0.61 (0.02)	0.59 (0.02)	0.61 (0.02)	0.63 (0.02)	0.67 (0.02)	0.70 (0.02)
T_nontarget	--	0.67 (0.02)	0.67 (0.02)	0.62 (0.02)	0.64 (0.03)	0.63 (0.02)	0.66 (0.02)	0.67 (0.03)	0.67 (0.02)	0.67 (0.03)	0.70 (0.02)
T_target	--	0.90 (0.01)	0.89 (0.01)	0.84 (0.02)	0.87 (0.01)	0.85 (0.02)	0.89 (0.01)	0.87 (0.02)	0.89 (0.01)	0.90 (0.01)	0.91 (0.01)

Note. Values are M (SE). T_foil = studied foil items; T_nontarget = studied but not initially tested;

T_target = studied and initially tested. Pos 0 represents foils (unstudied items).

Table 2

E1 Final Test Within List: Hits and Correct Rejections by Position Type and Position

Initial Study Position												
Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos	
10 Pos 11	Pos 12	Pos 13	Pos 14	Pos 15	Pos 16	Pos 17	Pos 18	Pos 19	Pos 20			
FOIL	0.83 (0.01)	--	--	--	--	--	--	--	--	--	--	--
--	--	--	--	--	--	--	--	--	--	--	--	--
T_foil	0.64 (0.01)	--	--	--	--	--	--	--	--	--	--	--
--	--	--	--	--	--	--	--	--	--	--	--	--
T_nontarget	-- 0.69 (0.02)	0.68 (0.02)	0.67 (0.02)	0.71 (0.02)	0.68 (0.02)	0.67 (0.02)	0.66 (0.02)	0.66 (0.02)	0.66 (0.02)	0.66 (0.02)	0.66 (0.02)	0.66 (0.02)
	0.66 (0.02)	0.64 (0.02)	0.61 (0.02)	0.64 (0.02)	0.66 (0.02)	0.65 (0.02)	0.64 (0.02)	0.65 (0.02)	0.66 (0.02)	0.66 (0.02)	0.70 (0.02)	0.67 (0.02)
T_target	-- 0.91 (0.01)	0.87 (0.02)	0.86 (0.02)	0.90 (0.01)	0.88 (0.01)	0.86 (0.02)	0.88 (0.01)	0.90 (0.01)	0.87 (0.02)	0.87 (0.02)	0.87 (0.02)	0.87 (0.02)
	0.87 (0.01)	0.86 (0.02)	0.89 (0.01)	0.89 (0.01)	0.87 (0.02)	0.87 (0.02)	0.89 (0.01)	0.89 (0.01)	0.87 (0.02)	0.87 (0.02)	0.87 (0.02)	0.87 (0.02)
Initial Test Position												
Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos	
10 Pos 11	Pos 12	Pos 13	Pos 14	Pos 15	Pos 16	Pos 17	Pos 18	Pos 19	Pos 20			
FOIL	0.83 (0.01)	--	--	--	--	--	--	--	--	--	--	--
--	--	--	--	--	--	--	--	--	--	--	--	--
T_foil	-- 0.51 (0.02)	0.55 (0.02)	0.60 (0.02)	0.64 (0.02)	0.65 (0.02)	0.61 (0.02)	0.63 (0.02)	0.67 (0.02)	0.65 (0.02)	0.65 (0.02)	0.68 (0.02)	0.71 (0.02)
	0.65 (0.02)	0.69 (0.02)	0.69 (0.02)	0.66 (0.02)	0.64 (0.02)	0.63 (0.02)	0.67 (0.02)	0.67 (0.02)	0.67 (0.02)	0.67 (0.02)	0.68 (0.02)	0.71 (0.02)

T_nontarget	0.66 (0.01)	--	--	--	--	--	--	--	--	--	--	--	--
--	--	--	--	--	--	--	--	--	--	--	--	--	--
T_target	--	0.80 (0.02)	0.85 (0.02)	0.85 (0.02)	0.87 (0.02)	0.88 (0.01)	0.87 (0.02)	0.88 (0.01)	0.88 (0.02)	0.88			
	(0.02)	0.89 (0.02)	0.88 (0.02)	0.89 (0.01)	0.88 (0.01)	0.87 (0.02)	0.90 (0.01)	0.90 (0.01)	0.90 (0.01)	0.90 (0.01)	0.88 (0.01)		
	0.89 (0.01)												

Note. Values are M (SE). T_foil = neither studied nor tested (studied foils); T_target = studied and tested;

T_nontarget = studied only. FOIL = unstudied items (new foils).

Table 3

E1 Initial Between List: Hits and Correct Rejections by List Number

Probe Type	List 1	List 2	List 3	List 4	List 5	List 6	List 7	List 8	List 9	List 10
Foil	0.93 (0.01)	0.94 (0.01)	0.94 (0.01)	0.93 (0.01)	0.92 (0.01)	0.93 (0.01)	0.94 (0.01)	0.94 (0.01)	0.93 (0.01)	0.93 (0.01)
Target	0.94 (0.01)	0.91 (0.01)	0.86 (0.01)	0.86 (0.01)	0.84 (0.01)	0.85 (0.01)	0.83 (0.01)	0.83 (0.01)	0.83 (0.01)	0.85 (0.01)

Note. Values are M (SE). All conditions combined. Foil = studied foil items (correct rejections);

Target = studied target items (hits).

Table 4

E1 Initial Within List: Hits and Correct Rejections by Position Type and Position

Initial Study Position

Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10
Pos 11	Pos 12	Pos 13	Pos 14	Pos 15	Pos 16	Pos 17	Pos 18	Pos 19	Pos 20		
Foil	0.93 (0.01)	--	--	--	--	--	--	--	--	--	--
--	--	--	--	--	--	--	--	--	--	--	--
Target	--	0.93 (0.01)	0.87 (0.01)	0.88 (0.01)	0.87 (0.01)	0.87 (0.01)	0.84 (0.01)	0.86 (0.01)	0.86 (0.01)	0.83 (0.01)	0.84 (0.01)
	0.85 (0.01)	0.87 (0.01)	0.84 (0.02)	0.85 (0.01)	0.84 (0.02)	0.85 (0.01)	0.86 (0.01)	0.87 (0.01)	0.87 (0.01)	0.86 (0.01)	

Initial Test Position

Probe Type	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10
Pos 11	Pos 12	Pos 13	Pos 14	Pos 15	Pos 16	Pos 17	Pos 18	Pos 19	Pos 20		
Average	--	0.86 (0.01)	0.89 (0.01)	0.88 (0.01)	0.89 (0.01)	0.90 (0.01)	0.90 (0.01)	0.89 (0.01)	0.91 (0.01)	0.90 (0.01)	
	0.91 (0.01)	0.90 (0.01)	0.90 (0.01)	0.89 (0.01)	0.89 (0.01)	0.91 (0.01)	0.90 (0.01)	0.89 (0.01)	0.90 (0.01)	0.90 (0.01)	0.90 (0.01)
Foil	--	0.95 (0.01)	0.95 (0.01)	0.93 (0.01)	0.95 (0.01)	0.95 (0.01)	0.94 (0.01)	0.94 (0.01)	0.95 (0.01)	0.94 (0.01)	0.93 (0.01)
	0.93 (0.01)	0.93 (0.01)	0.92 (0.01)	0.92 (0.01)	0.93 (0.01)	0.93 (0.01)	0.91 (0.01)	0.93 (0.01)	0.92 (0.01)	0.92 (0.01)	
Target	--	0.76 (0.02)	0.84 (0.02)	0.82 (0.02)	0.84 (0.01)	0.85 (0.01)	0.86 (0.01)	0.86 (0.01)	0.86 (0.01)	0.87 (0.01)	0.88 (0.01)
	0.88 (0.01)	0.87 (0.01)	0.86 (0.01)	0.87 (0.01)	0.89 (0.01)	0.87 (0.01)	0.88 (0.01)	0.88 (0.01)	0.87 (0.01)	0.89 (0.01)	

Note. Values are M (SE). Foil = studied foil items (correct rejections); Target = studied target

items (hits). Average shown for Initial Test Position only (mean across foil and target).